集成电路原理与设计

2. 工艺流程

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Syllabus

课数	内容	课数	内容
1	导论	9	放大器
2	工艺流程	10	放大器
3	器件模型	11	逻辑门
4	模拟基本单元	12	组合逻辑
5	电流镜与基准	13	时序逻辑
6	单级放大器	14	加法器/乘法器
7	放大器	15	集成电路专题讲座一
8	课堂测验	16	集成电路专题讲座二

Goal of this chapter

- Overview of Manufacturing Process
- Layout
- Appendix: Design Rules

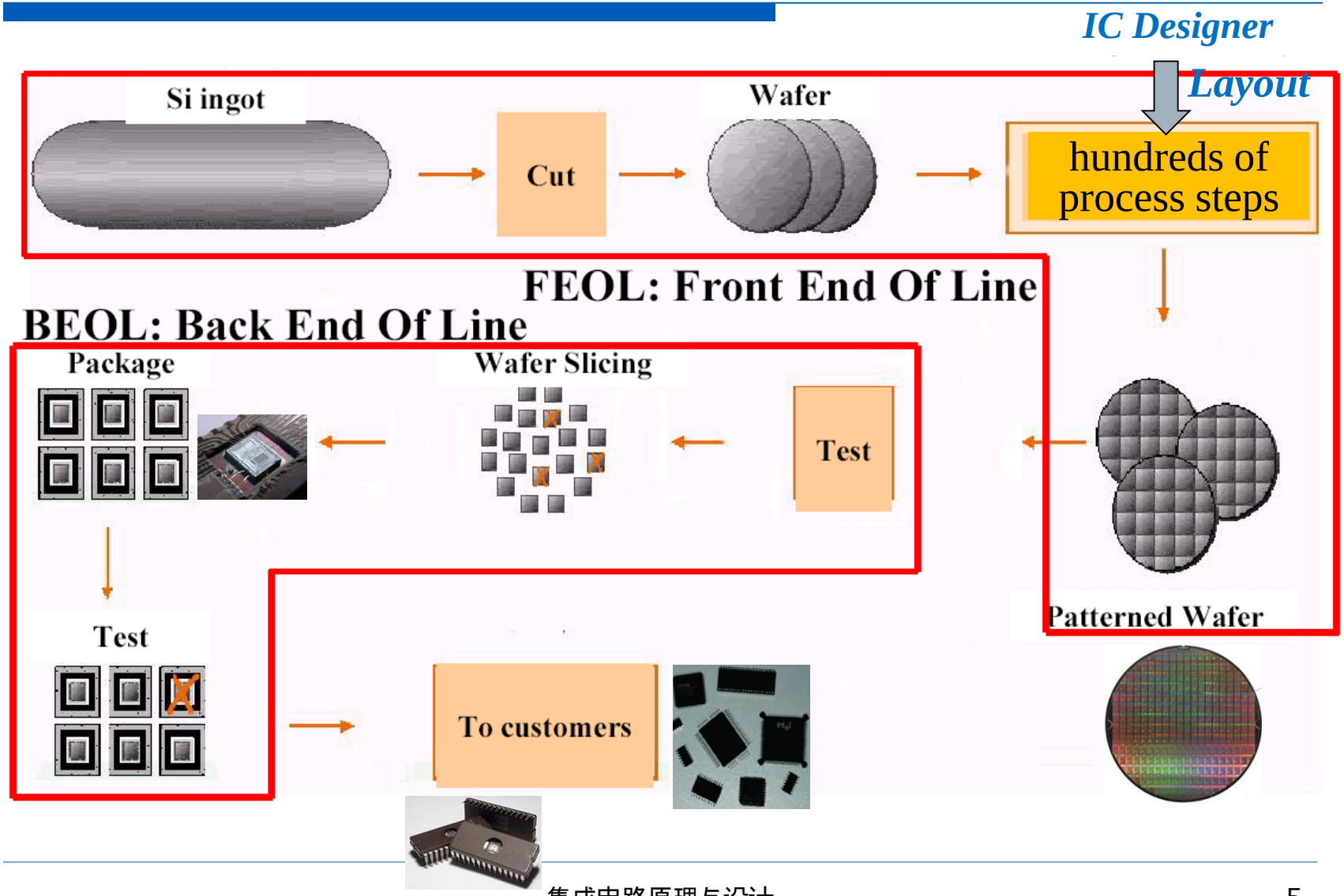
Reference:
Chapter 18, <Design of Analog CMOS Integrated Circuits>Razavi著

Outline

- Basic Semiconductor Fabrication Processes
- CMOS Fabrication Steps
- Layout

- Appendix: Design Rules

Chip fabrication process - Manufacturing industry



Top Foundries: 2024Q4

4Q24全球前十大晶圆代工业者营收排名 (单位：百万美元)

Ranking	Company	Revenue			Market Share	
		4Q24	3Q24	QoQ	4Q24	3Q24
1	台积电(TSMC)	26,854	23,527	14.1%	67.1%	64.7%
2	三星(Samsung)	3,260	3,305	-1.4%	8.1%	9.1%
3	中芯国际(SMIC)	2,207	2,171	1.7%	5.5%	6.0%
4	联电(UMC)	1,867	1,873	-0.3%	4.7%	5.1%
5	格芯(GlobalFoundries)	1,830	1,739	5.2%	4.6%	4.8%
6	华虹集团 (Huahong Group)	1,042	982	6.1%	2.6%	2.7%
7	高塔半导体(Tower)	387	371	4.5%	1.0%	1.0%
8	世界先进(VIS)	357	366	-2.3%	0.9%	1.0%
9	合肥晶合(Nexchip)	344	332	3.7%	0.9%	0.9%
10	力积电(PSMC)	333	336	-0.7%	0.8%	0.9%
Total of Top 10		38,482	35,001	9.9%	96%	96%

备注1：3Q24--1美元兑换1,355韩元；1美元兑换32.3台币
备注2：4Q24--1美元兑换1,399韩元；1美元兑换32.3台币
备注3：三星仅计入晶圆代工事业部之营收
备注4：力积电仅计入记忆体与逻辑晶圆代工营收
备注5：华虹集团包含华虹宏力及上海华力
Source: TrendForce, March 2025

TRENDFORCE

知乎 @卓乎

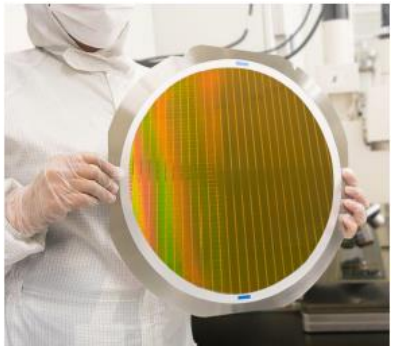
Fabrication

- Chips are built in huge factories called ***Fabs***
- Contain clean rooms as large as football fields











Courtesy of International
Business Machines Corporation.
Unauthorized use not permitted.

Outline

- **Basic Fabrication Semiconductor Processes**
  - **Some Recurring Process Steps:**

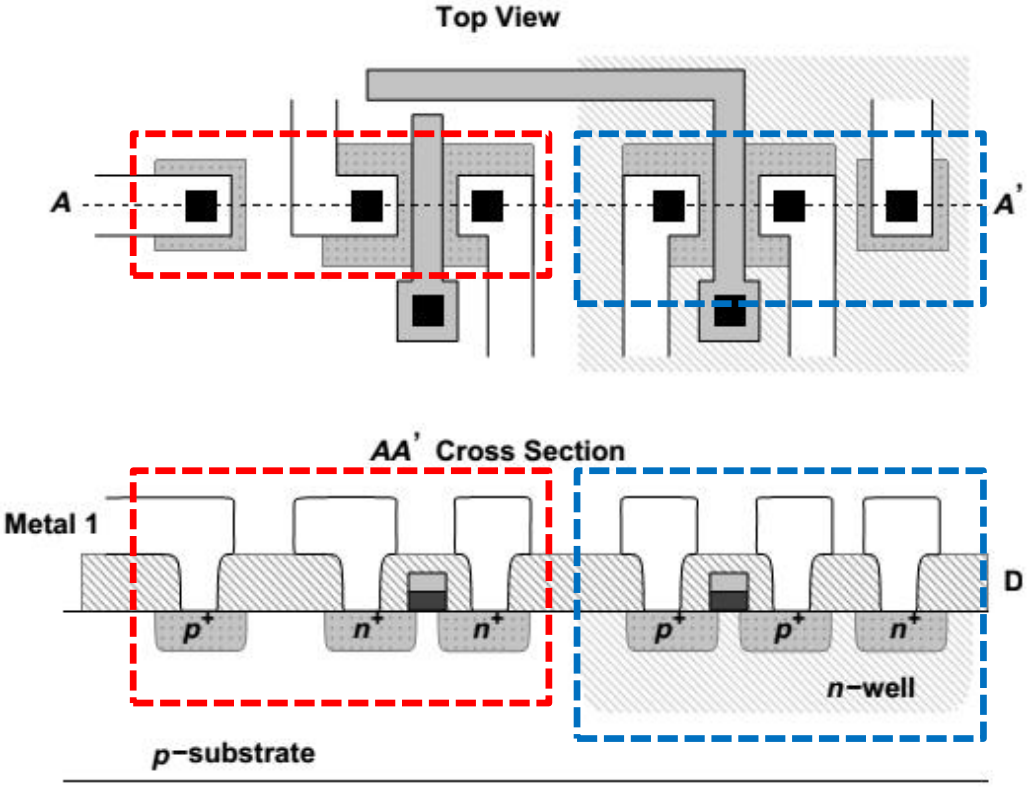
    Oxidation, Diffusion, Ion implantation, Deposition, Etching, Planarization

  - **Potolithography**
- **CMOS Fabrication Steps**
- **Layout**
- **Appendix: Design Rules**

Basic structure of NMOS/PMOS

- a p-type substrate (wafer) serves with n-wells
- source/drain regions, gate dielectric, polysilicon
- n-well and substrate ties
- metal interconnects



Predict the required processing steps

P-sub, N-well

CMOS process flow

Predict the required processing steps

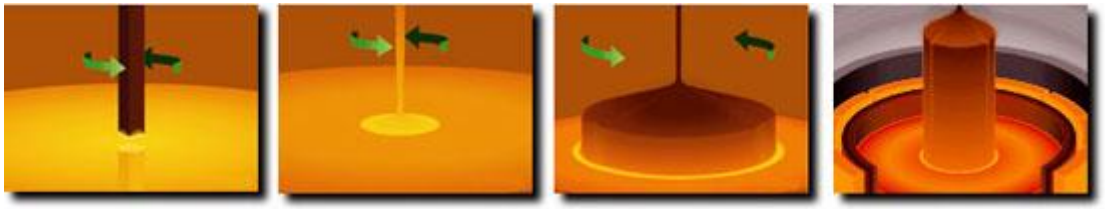
How are(1) various regions defined so accurately;
(2) the n-wells and S/D regions built;
(3) the gate oxide and polysilicon fabricated;
(4) the gate oxide and polysilicon aligned with the S/D regions;
(5) the contact windows created;
(6) the metal interconnect layers deposited ……

- Modern CMOS technologies involve **hundreds processing steps**
- The sequence as a combination of the following operations:
  - **Wafer**: produce the proper type of substrate
  - **Photolithography**: precisely define each region
  - **Oxidation, deposition** and **ion implantation:** add materials to the wafer
  - **Etching:** remove materials from the wafer

Repetitive!

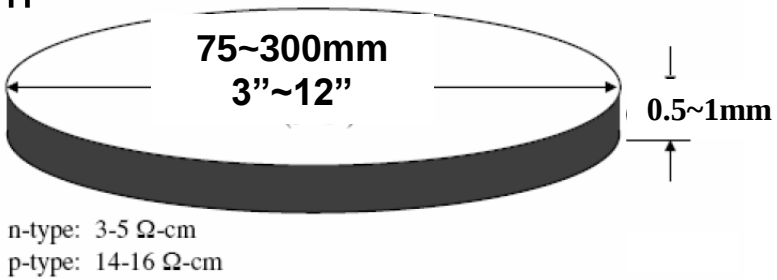
Silicon Wafer

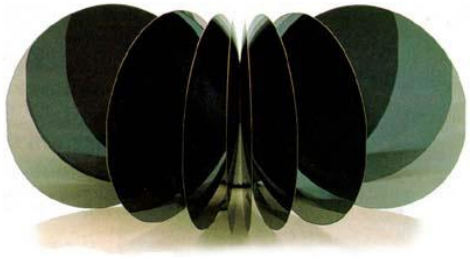
- Single-crystal silicon body with small number of “defects” 缺陷
  - dopants -> the desired resistivity
  - Czochralski process(切克劳斯基法) **25mm/h**





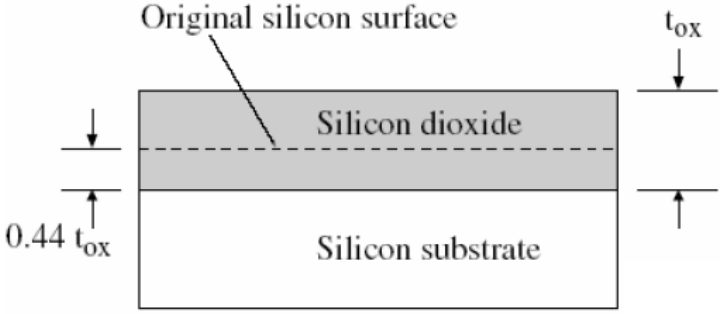
- Wafer
  - The wafers are polished and chemically etched
    Remove damages on the surface created during slicing
  - Length: ~1m





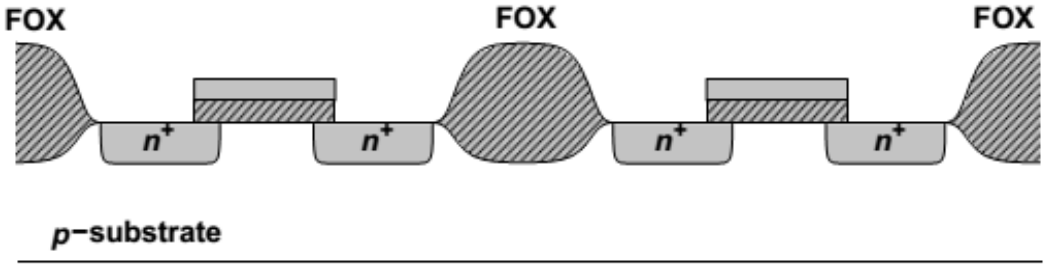
Oxidation 氧化

Oxidation is the process by which a layer of silicon dioxide (SiO_2) is formed on the surface of silicon wafer



- 生长方式
  - 干法（薄氧100~1000 Å）
  - 湿法（厚氧）
- 生长的温度：700~1100 ℃

- Gate oxide: gate dielectric t_{ox}<50A
- FOX (Field oxide场氧): in areas between the devices ~10,000A
- Isolation between two layers
- Protective coating-- protect the underlying material from contamination



Ion implantation离子注入

Dopant!

Ion implantation is the process by which ions of a particular dopant (impurity) are accelerated by an electric field to a high velocity and physically lodge within the semiconductor material.

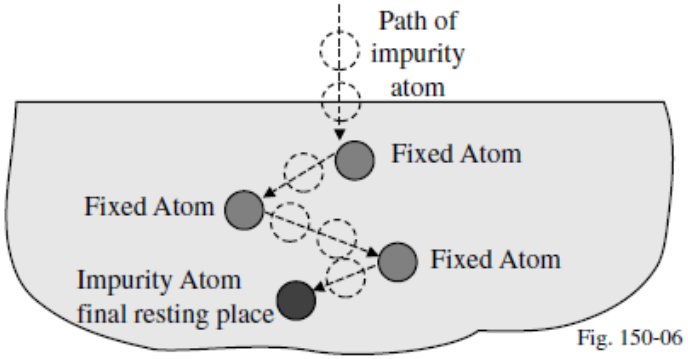
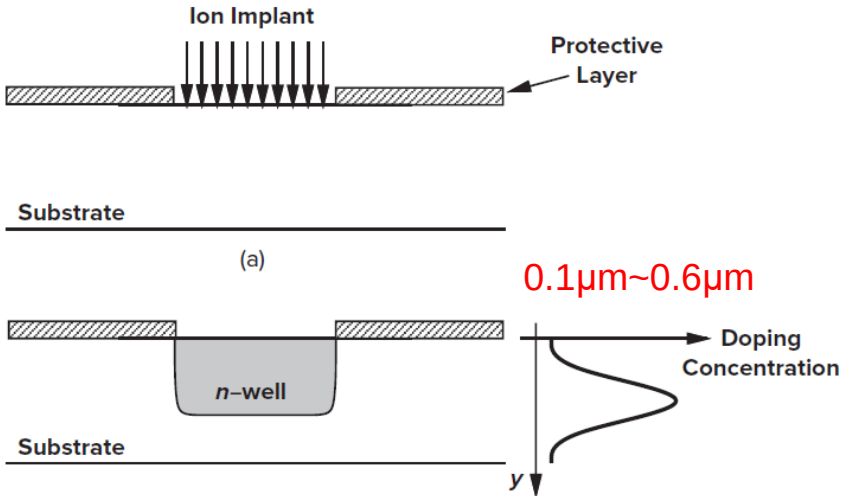


Fig. 150-06



- the profile of the implanted impurities
  - 掺杂的深度和浓度可控：±5%
  - 重复性好；Be titled by 7~9°
- Annealing(退火): 800℃

高速的离子注入会对半导体晶格产生破坏，使注入电子留在电不活动区。

⇒ 晶格再次形成，同时杂质扩散向各个方向展宽

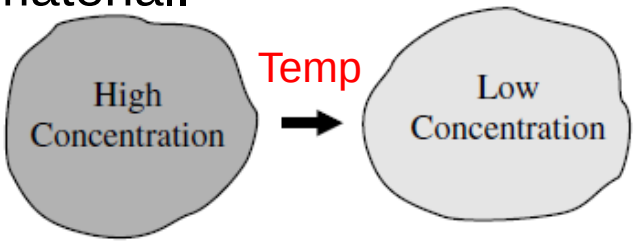
所有注入完成后，对wafer退火一次

Diffusion扩散

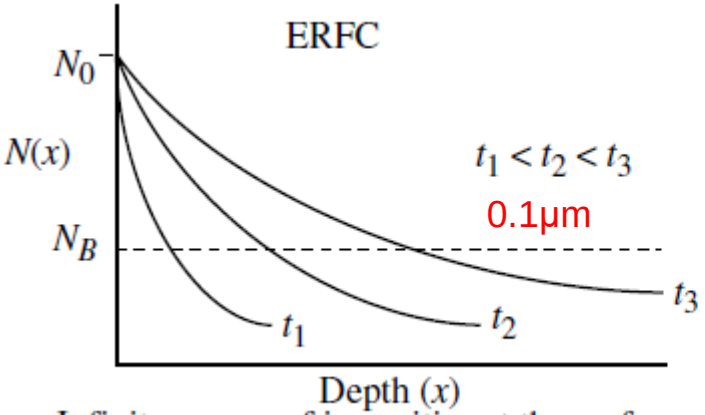
Dopant!

Diffusion in semiconductor is the movement of impurity atoms at the surface of the material into the bulk of the material.

- 800~1400℃

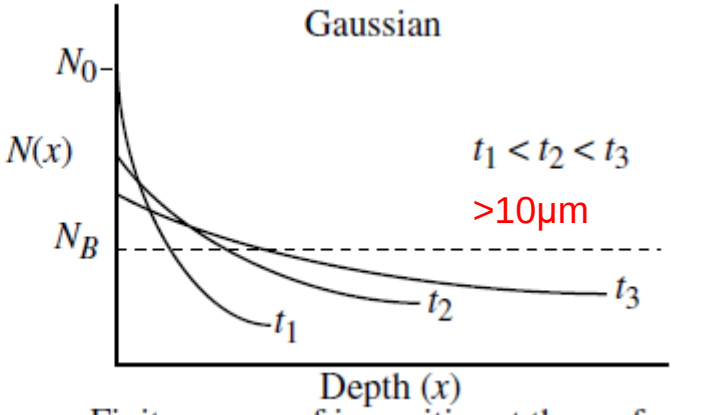


- The concentration profile of the impurity is a function of the concentration of the impurity at the surface and the time.



Infinite source of impurities at the surface.

表面无限杂质源（预淀积）



Finite source of impurities at the surface.

表面无杂质源（再分布）

Ion implantation vs. Diffusion

Objective：insert impurities into the semiconductor



The advantage of the ion implantation:

- The accurate control of doping (±5%) and good reproducibility

  Adjust the threshold of MOS, Channel stopper, Create precise resistor
- Implantation allows control the profile of the implanted impurities
- Room temperature process
- It is possible to implant through a thin layer

  The material to be implanted does not have to be exposed to contaminants during and after the implantation process

Deposition淀积/沉积

Deposition is the means by which films of various materials may be deposited on the silicon wafer

- Polysilicon
- Silicon Nitride(形成场氧和注入阻挡层时用来作为缓冲材料)
- Dielectric materials separating interconnect layers
- Metal layers serving as interconnects

Common methods:

- Evaporation蒸发
- Sputtering溅射 *Aluminum*
- CVD (chemical vapor deposition，化学气相沉积)

  Polysilicon, Si_3N_4

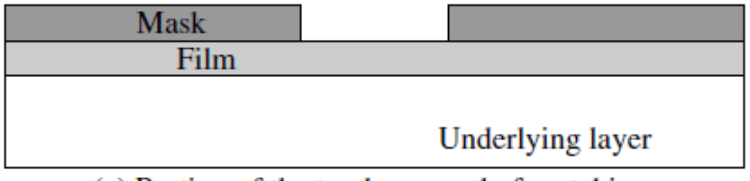


Note: Deposited material covers the entire wafer.

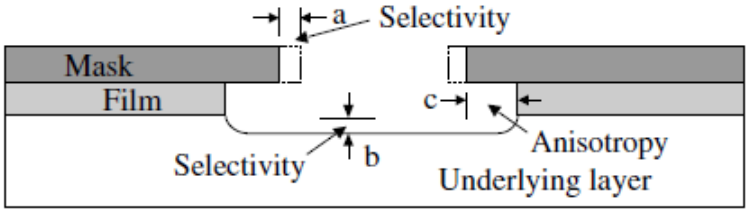
Etching刻蚀

Etching is the process of **removing exposed (unprotected)** of material.

- Selectivity(选择性)
- Anisotropy(各向异性)



(a) Portion of the top layer ready for etching.



(b) Horizontal etching and etching of underlying layer.

- **Wet etching**: placing the wafer in a chemical liquid (low precision)

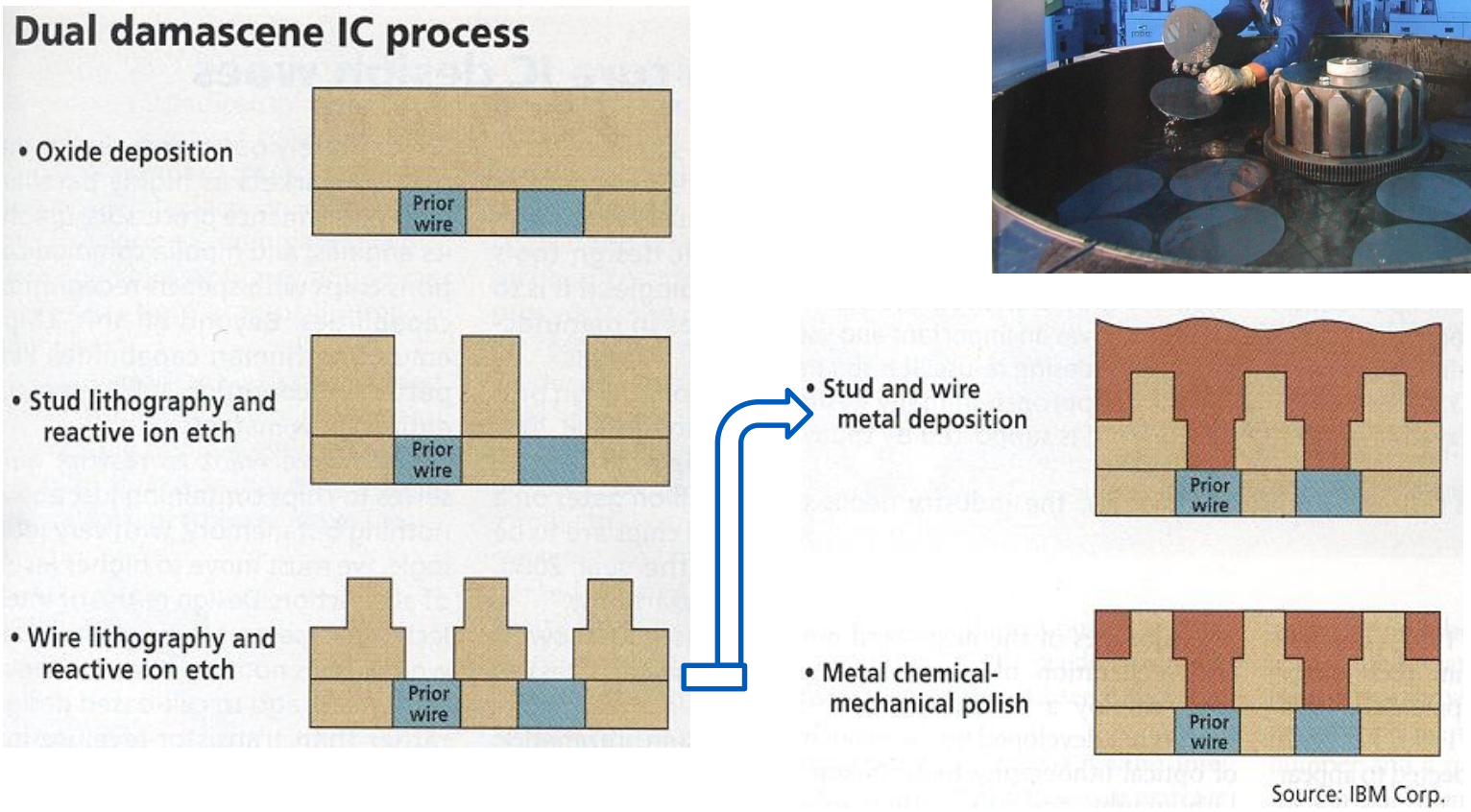
  HF=>SiO_2, H_3PO_4=>SiN_4, KOH=>Silicon, phosphoric acid mixture=>metal
- **Plasma etching**: bombarding the wafer with a plasma gas (high precision)
- **Reactive ion etching (RIE)**: where ions produced in a gas bombard the wafer

Planarization平坦化

Polishing the wafer

- **Chemical Mechanical Polishing (CMP)**



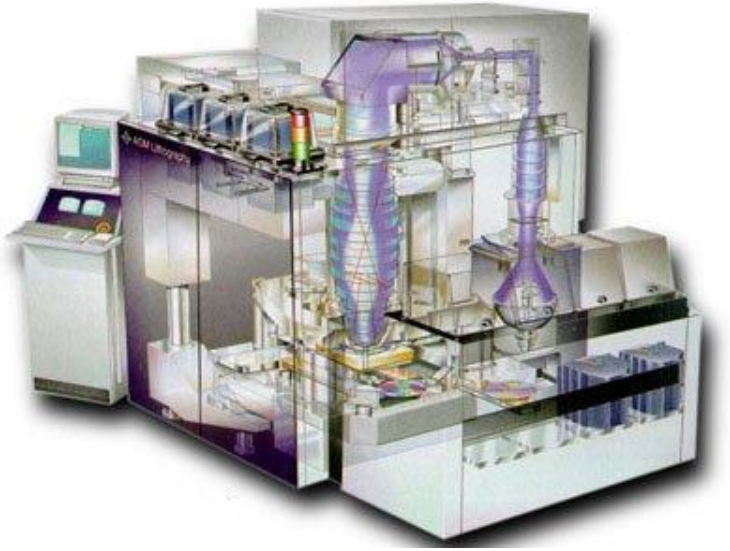


Photolithography光刻

-- Diffusion/ion implantation, etching are applied to selected parts of the silicon wafer ⇨ **Photolithography**

-- The process of transferring an image from a photomask or computer database to the wafer

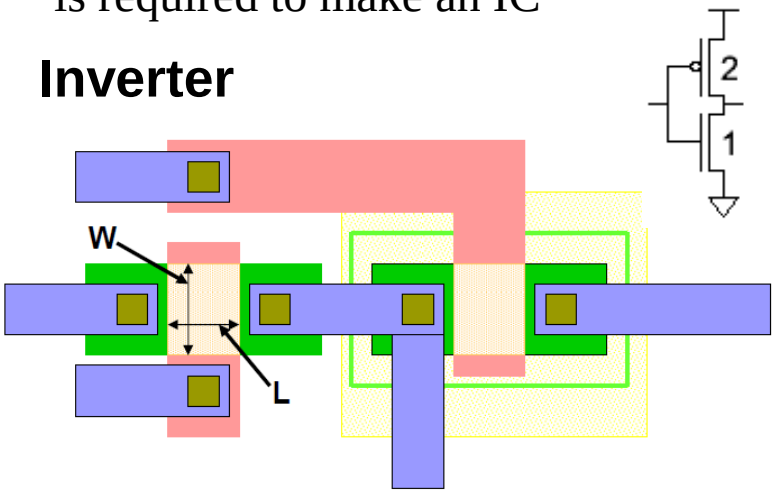
- Photomask(掩模): expose some areas of the photoresist to UV light while shielding the remainder
- Photoresist(光刻胶): an organic polymer whose characteristics can be altered when exposed to UV light
  - ✓ negative(负胶) – create a mask where patterns do not exist (where the photomask is transparent to UV light)
  - ✓ positive(正胶) – create a mask where patterns exist (where the photomask is opaque to UV light)
- 光源：电子束、UV ***DUV、EUV***

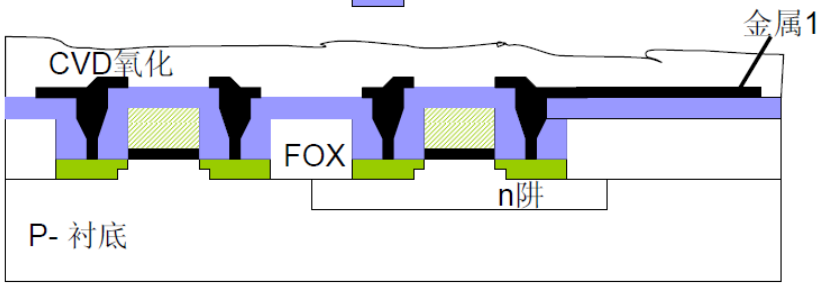


Photolithography: Mask掩模

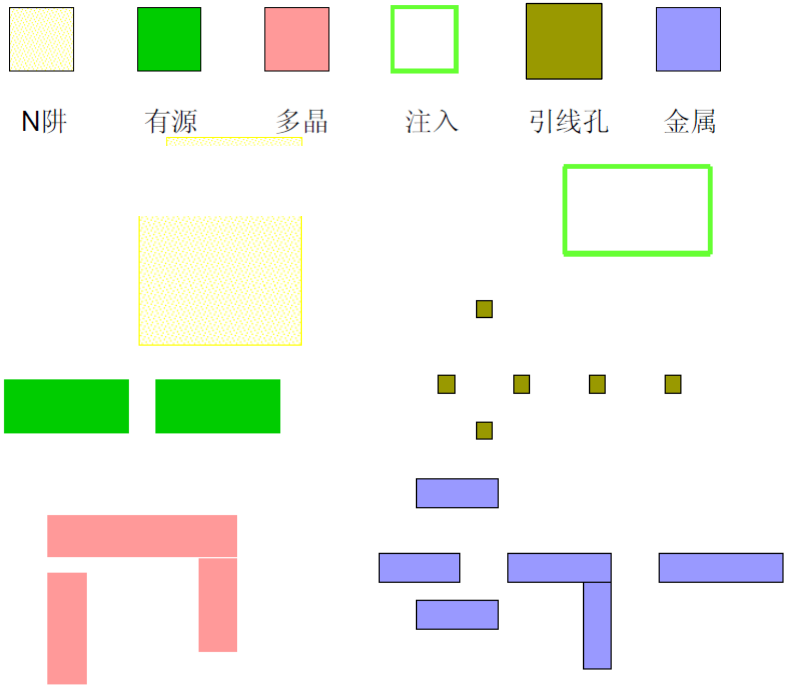
- ✓ Mask=a glass or quartz substrate + a thin layer of an opaque, hard-wearing material such as chromium
- ✓ Resolution ~ wavelength of the light
- ✓ One mask contains all the information to process one layer of chip, so a set of masks is required to make an IC

Inverter





Masks



Photolithography: steps(1)

- 步骤：涂胶-预烘-曝光(光刻版)-显影-坚膜-刻蚀-去胶

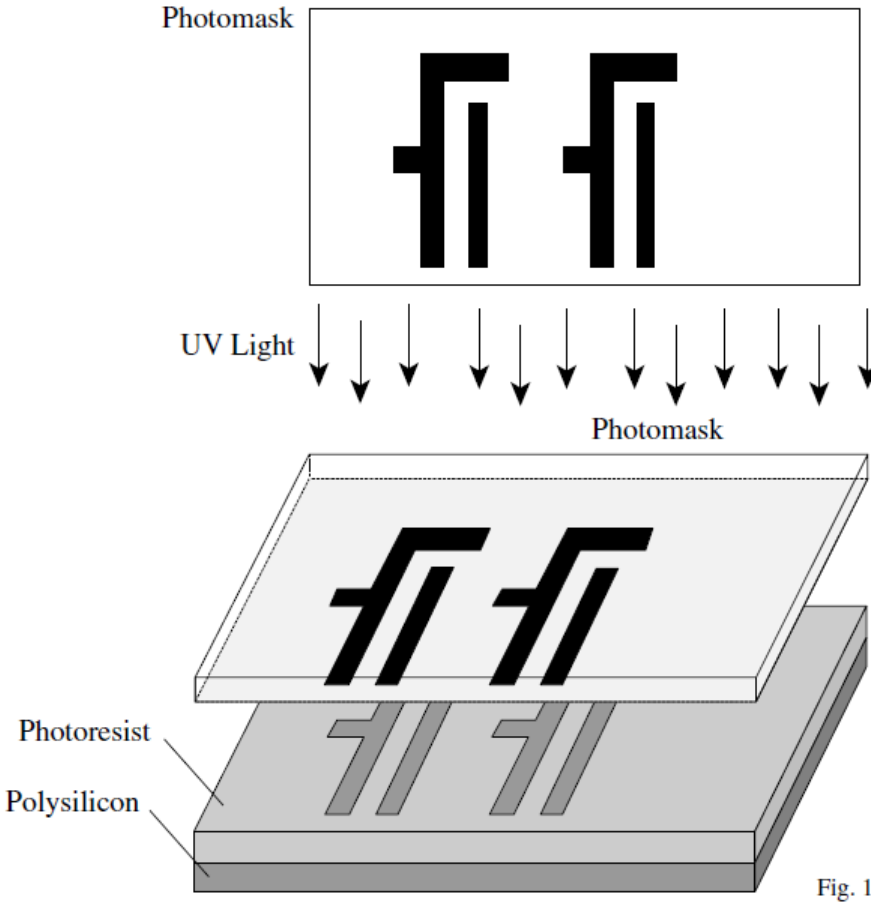
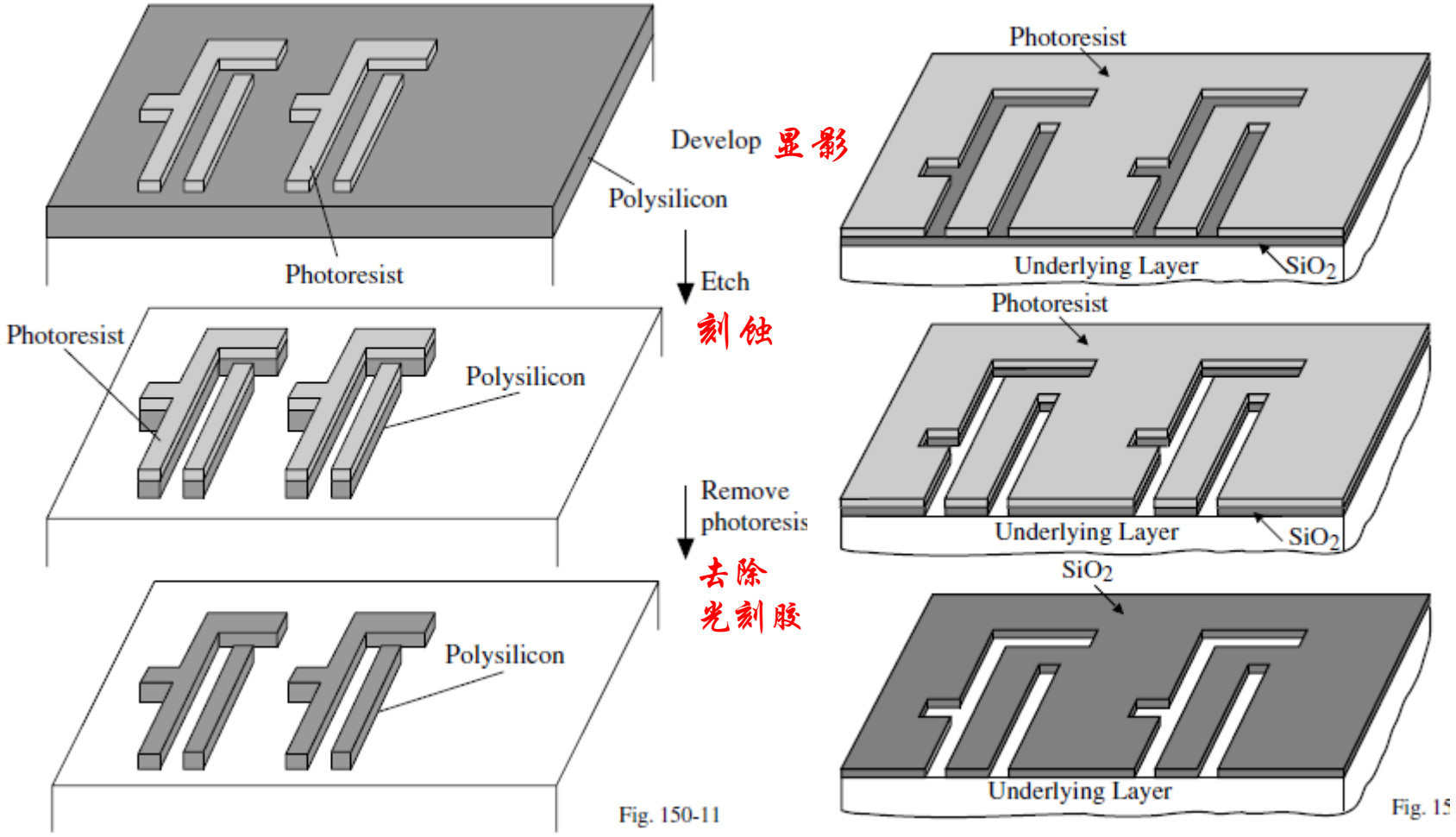


Fig. 1

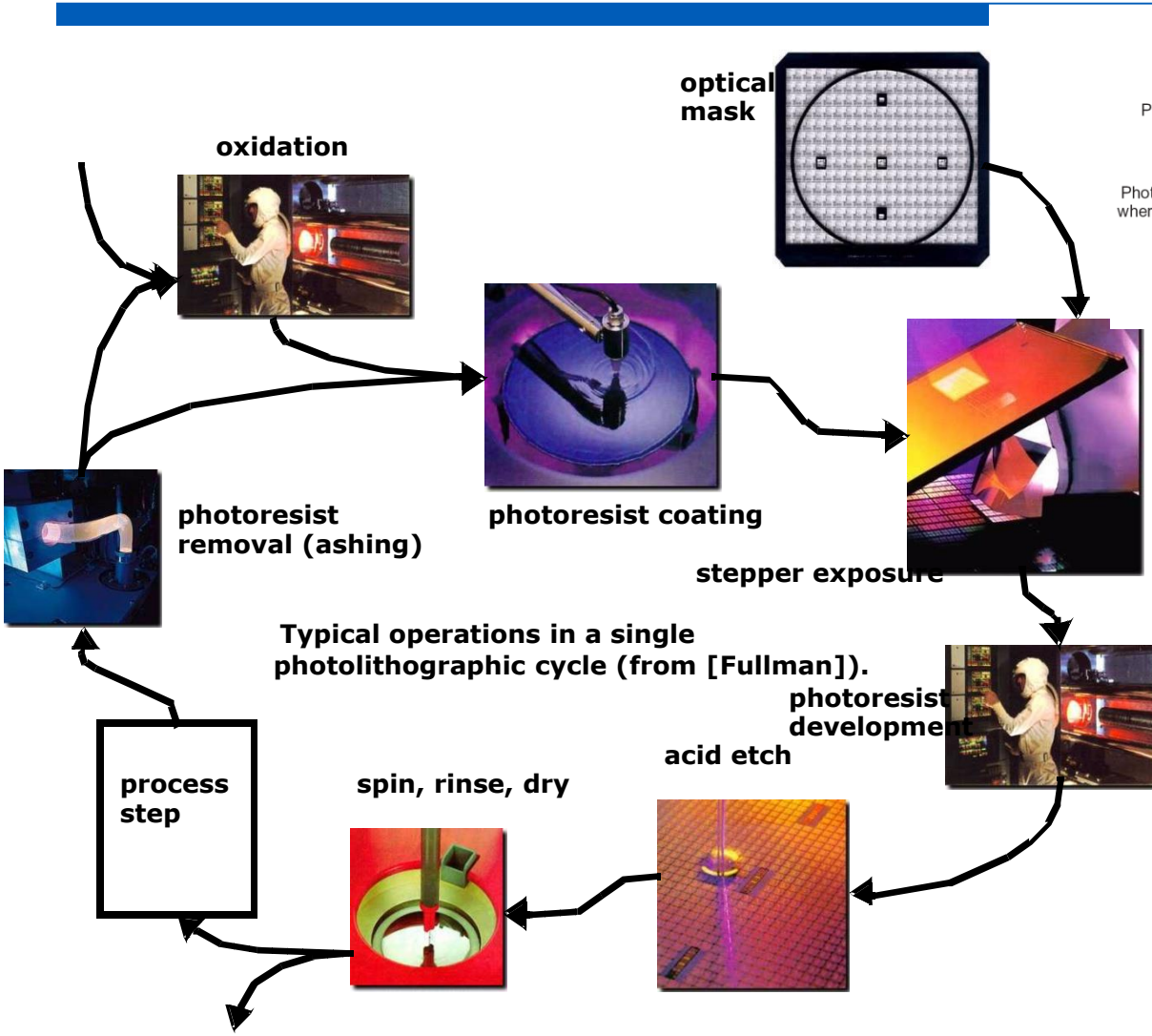
Photolithography: steps（2）



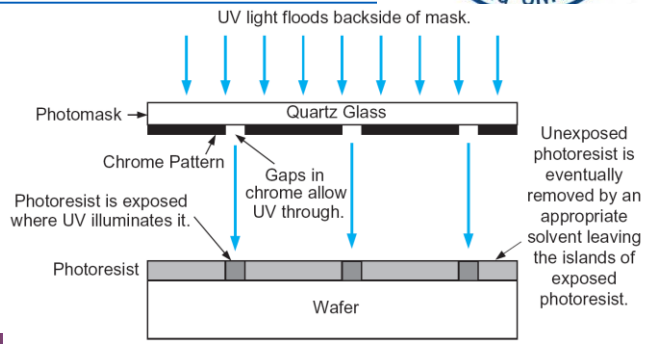
Positive photeresist

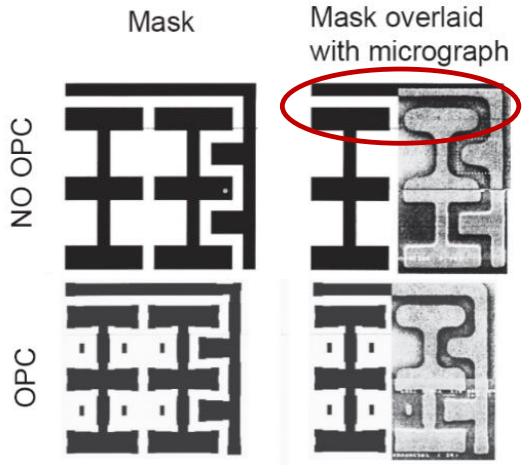
Negative photeresist

Appendix: Photo-Lithographic Process

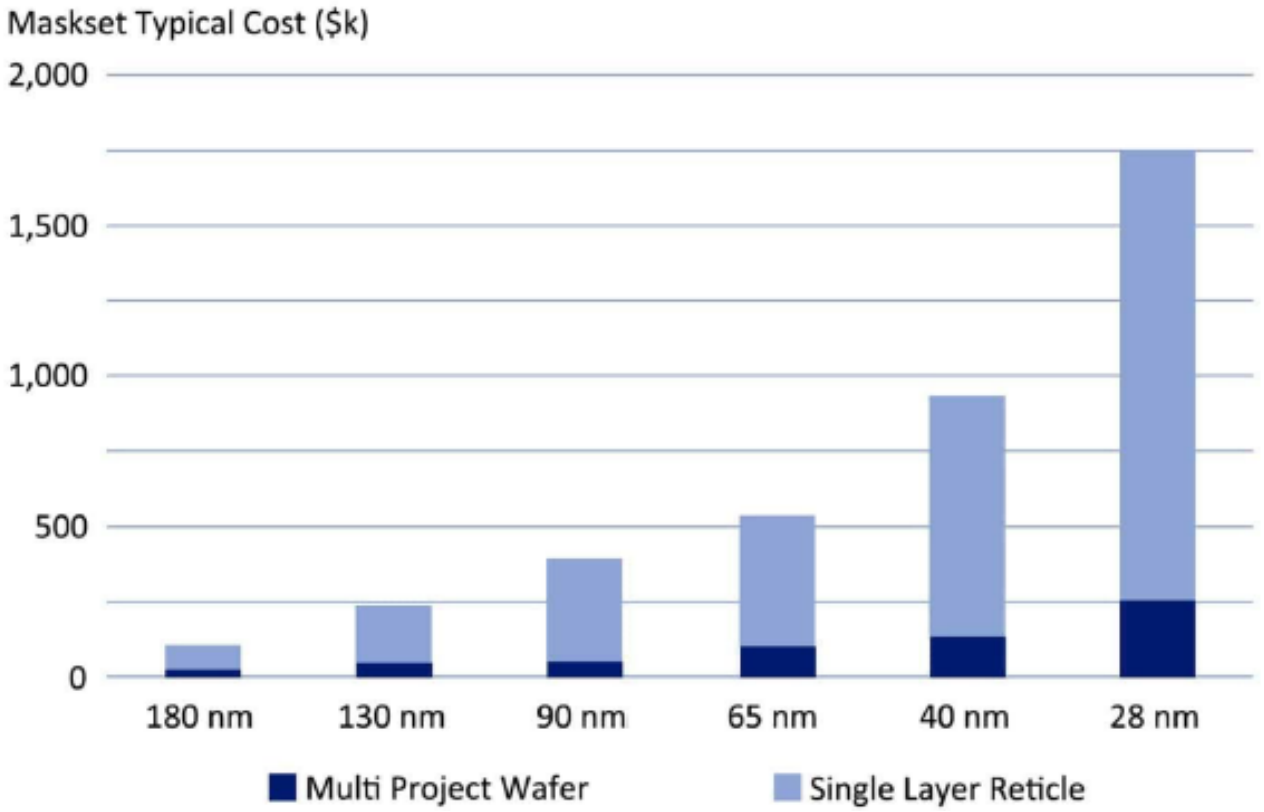


Typical operations in a single photolithographic cycle (from [Fullman]).





Appendix: Mask costs



MPW(Multi-Project Wafer): Share cost of mask tooling between multiple designs!

- **Prototyping**
- **Low volume production**

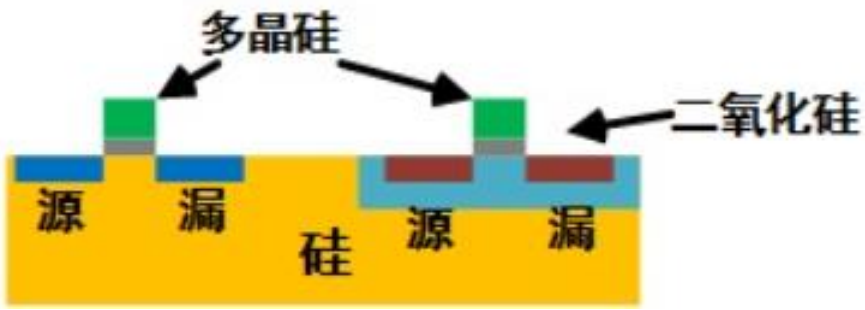
Outline

- Basic Semiconductor Fabrication Processes
- CMOS Fabrication Steps
  - Bulk Si vs. SOI Si
  - Planar MOS vs. FinFET
  - CMOS Process：N-well Silicon-gate CMOS
- Layout

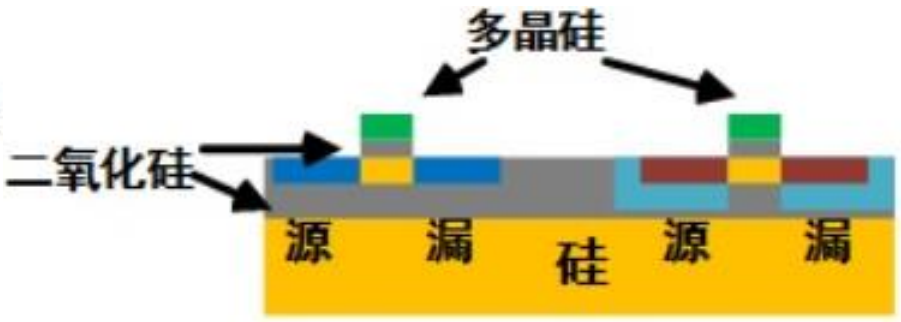
- Appendix: Design Rules

Bulk Si vs. SOI Si

- Bulk Si
  - Conventional process
  - Used by Intel (14nm-> FinFET).
- SOI (Silicon-on-Insulator) Si
  - Source/drain and substrate/bulk are separated by SiO_2
  - Used by IBM(->Global Foundry).



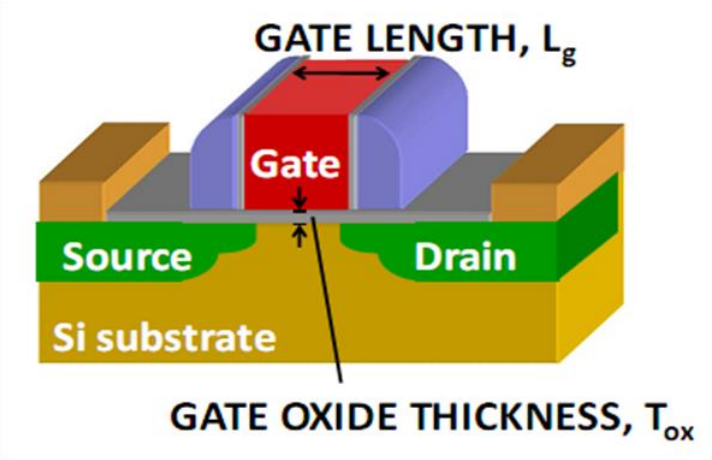
Bulk Si

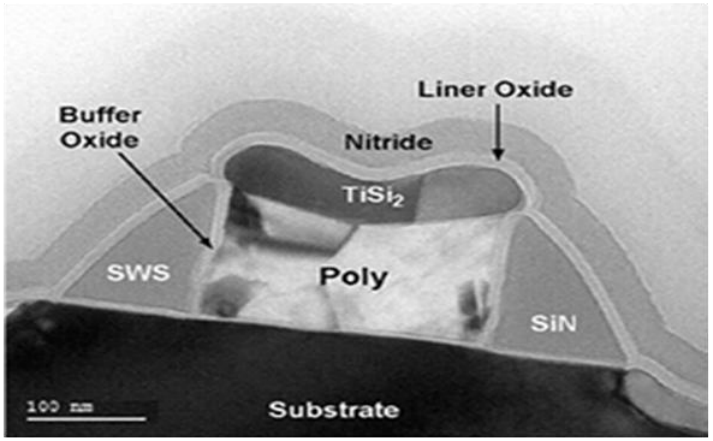


SOI Si

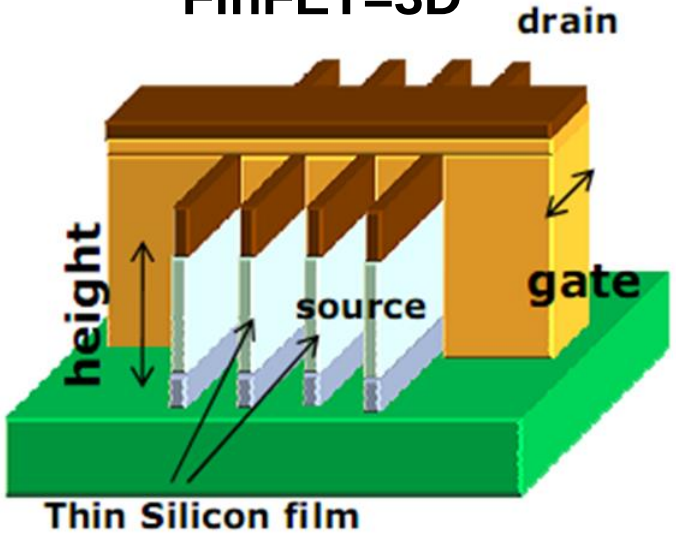
Planar MOS vs. FinFET

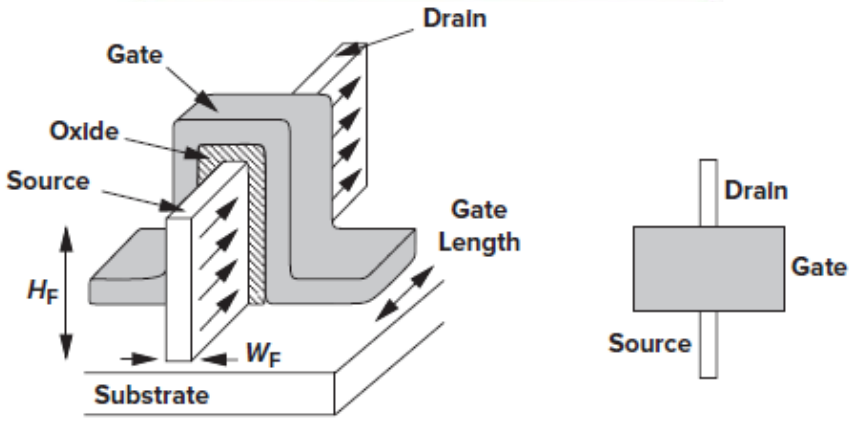
Planar MOS=2D





FinFET=3D



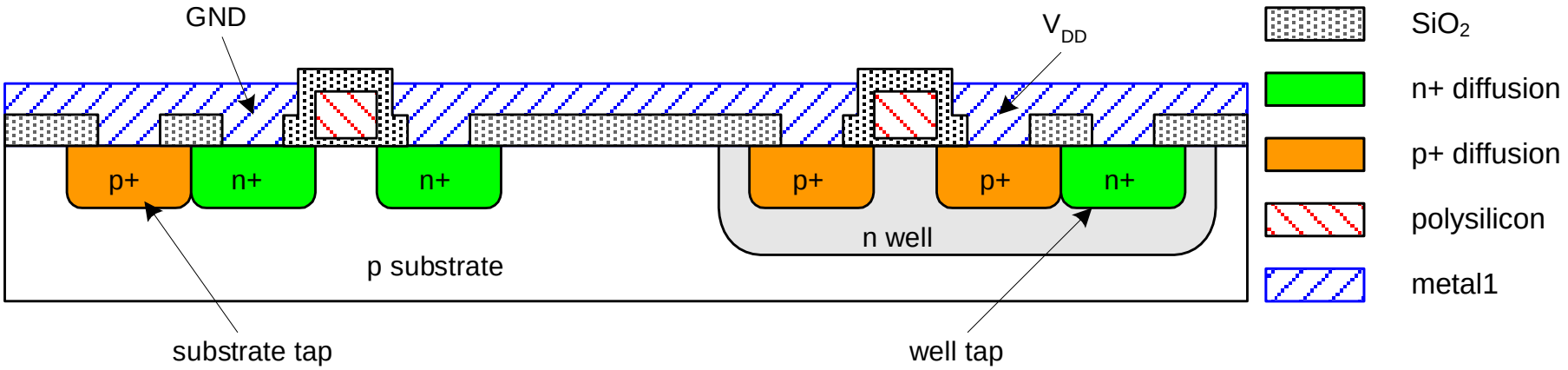


CMOS Process

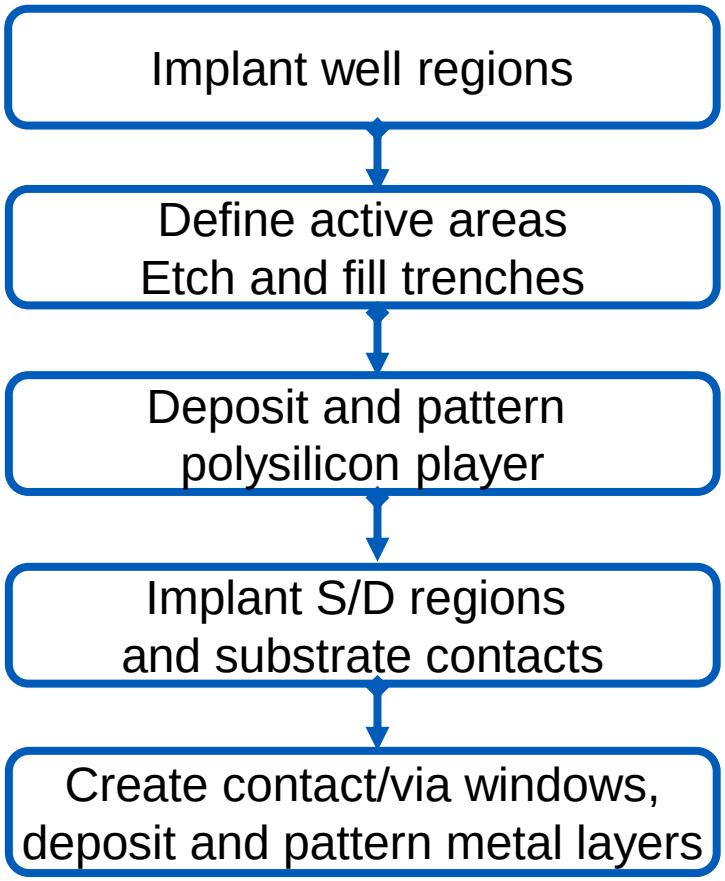
- CMOS transistors are fabricated on silicon wafer
- Lithography process similar to printing press
- On each step, different materials are deposited or etched
- Easiest to understand by viewing cross-section of wafer in a simplified manufacturing process

N-well Silicon-gate CMOS Fabrication

- Typically use p-type substrate for nMOS transistors
- Requires n-well for body of pMOS transistors
- Substrate must be tied to GND
- Metal to lightly-doped semiconductor forms poor connection called Shottky Diode
- Use heavily doped well and substrate contacts / taps

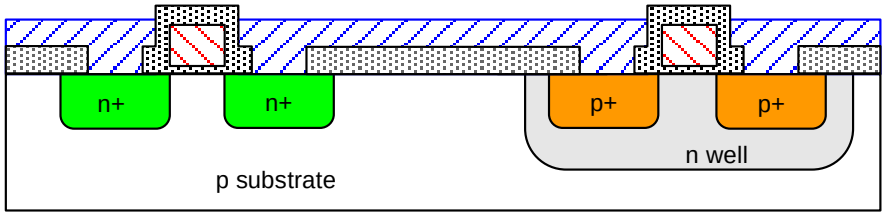


CMOS Process at a Glance

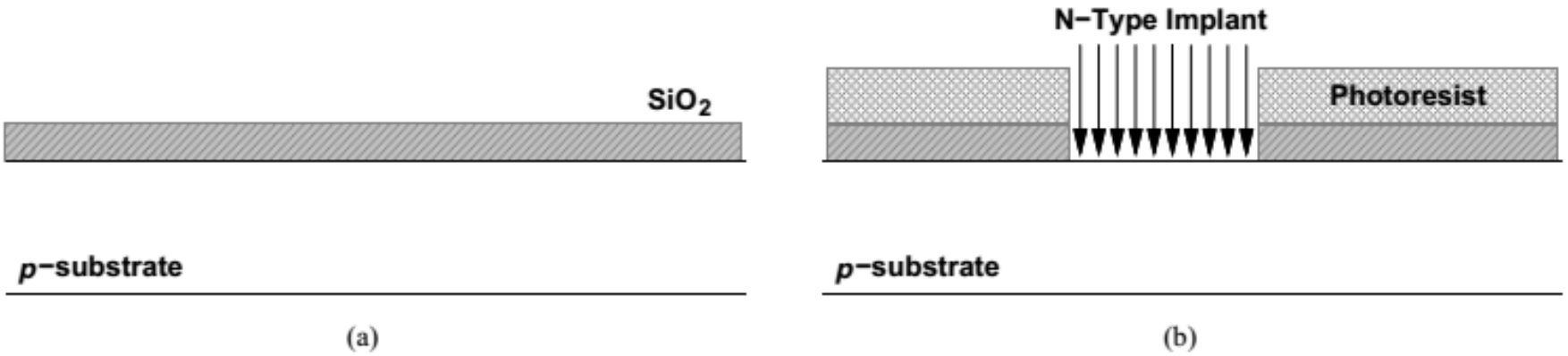


N-well Silicon-gate CMOS Fabrication

- 在P型衬底上形成N型Well(阱)
- 确定有源区(active)和场区(field)，场区隔离(isolation)
- 制备MOS管: 阈值电压调节，
  自对准结构，
  LDD(Light-doping drain)，。。。
- 元件互连
- 钝化



N-well Silicon-gate CMOS Fabrication (1)

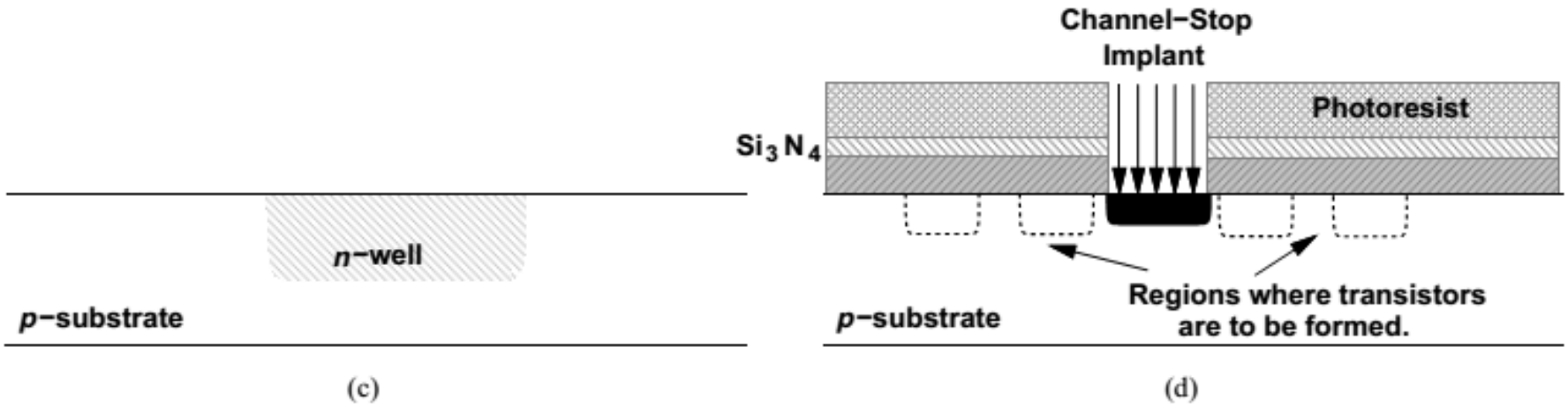


- p-type silicon wafer with ~1 mm thick
- cleaning and polishing
- grow a thin layer of SiO_2 (as a protective coating)

N-well mask: create the n-wells

- a lithography sequence:
  photoresist deposition
  exposure to UV light
  selective etching
- n-wells are implanted.

N-well Silicon-gate CMOS Fabrication (2)



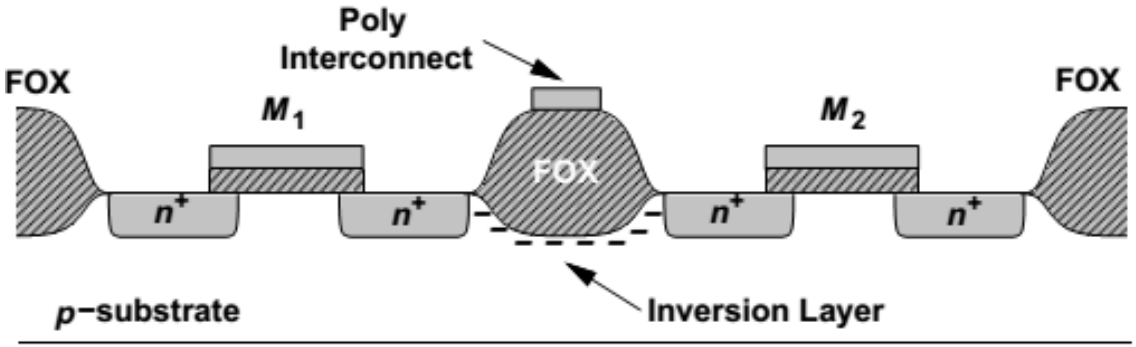
- remove the remaining photoresist and oxide layers

Active mask

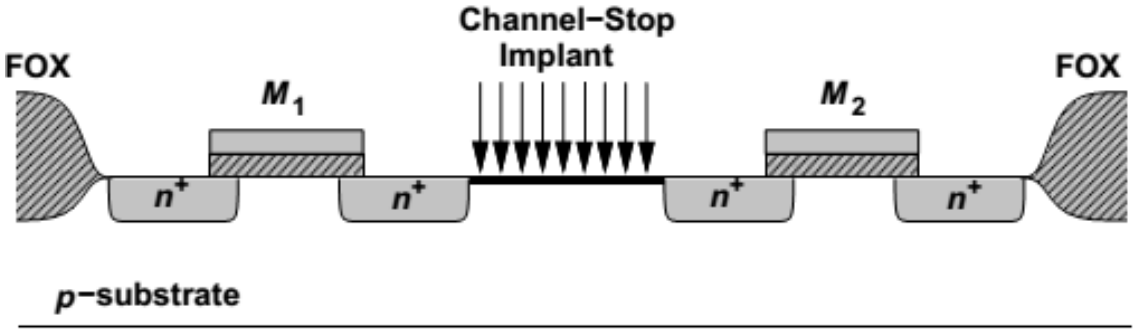
- SiO_2+ Si_3N_4+ positive photoresist
- **Channel-stop implant**

  remove the photoresist
  grow a thick oxide layer in the exposed silicon areas, producing the field oxide

Appendix: Channel-stop implant



(a)



(b)

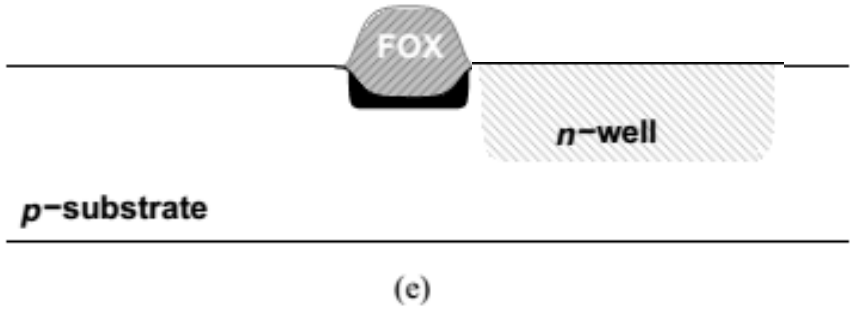
(a) Unwanted conduction due to inversion of field area, (b) channel-stop implant.

$$V_{TH} = \Phi_{MS} + 2\Phi_F + \frac{Q_{dep}}{C_{ox}}$$

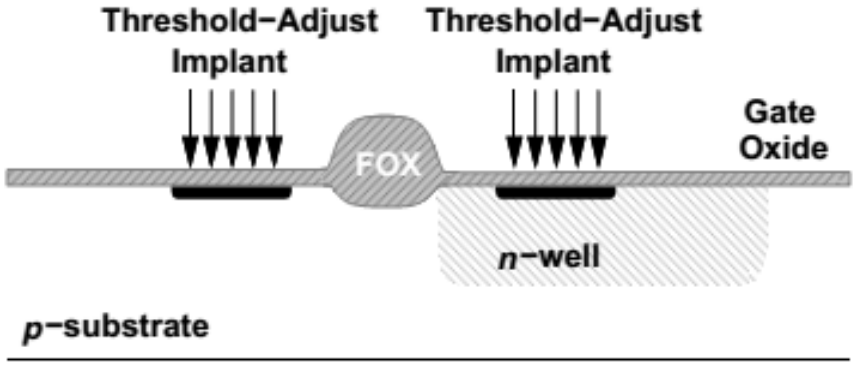
$$\Phi_F = (kT/q)\ln(N_{sub}/n_i)$$

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{\varepsilon_o \varepsilon_{sio_2}}{t_{ox}}$$

N-well Silicon-gate CMOS Fabrication (3)



(e)



(f)

- remove the protective nitride and oxide layers
  exposing all areas where transistors are to be formed
- FOX: field oxide场氧

- growth of the gate oxide:
  slow, low-pressure CVD
- threshold-adjust implant

Appendix: Isolation（1）

- Self-isolated
- LOCOS (LOCalized Oxidation of Silicon)
- Shallow Trench Isolation (STI)

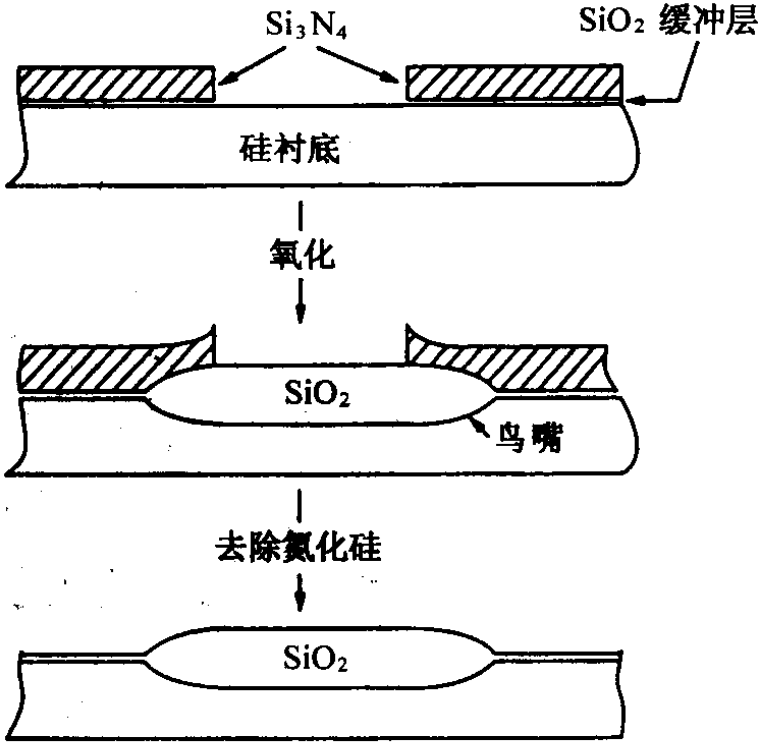
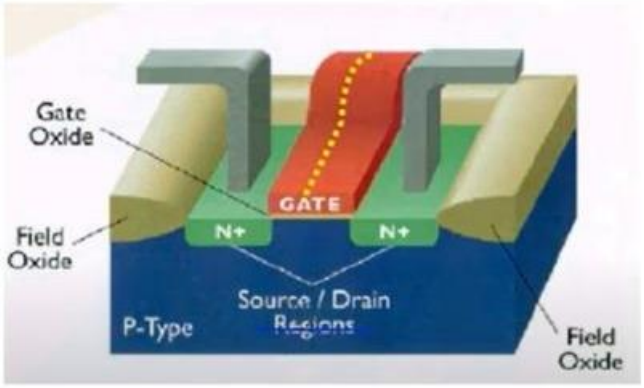


图 10.2　局部场氧化工艺步骤



3D figure

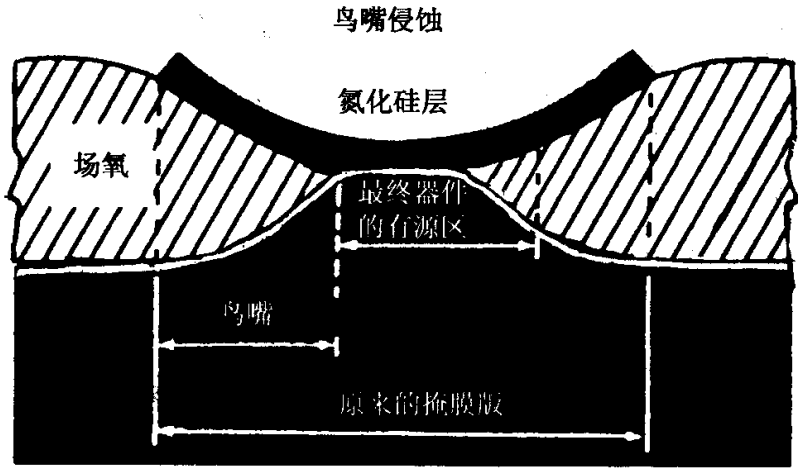


图 10.3　鸟嘴示意图

Appendix: Isolation（2）

- Shallow Trench Isolation (STI)

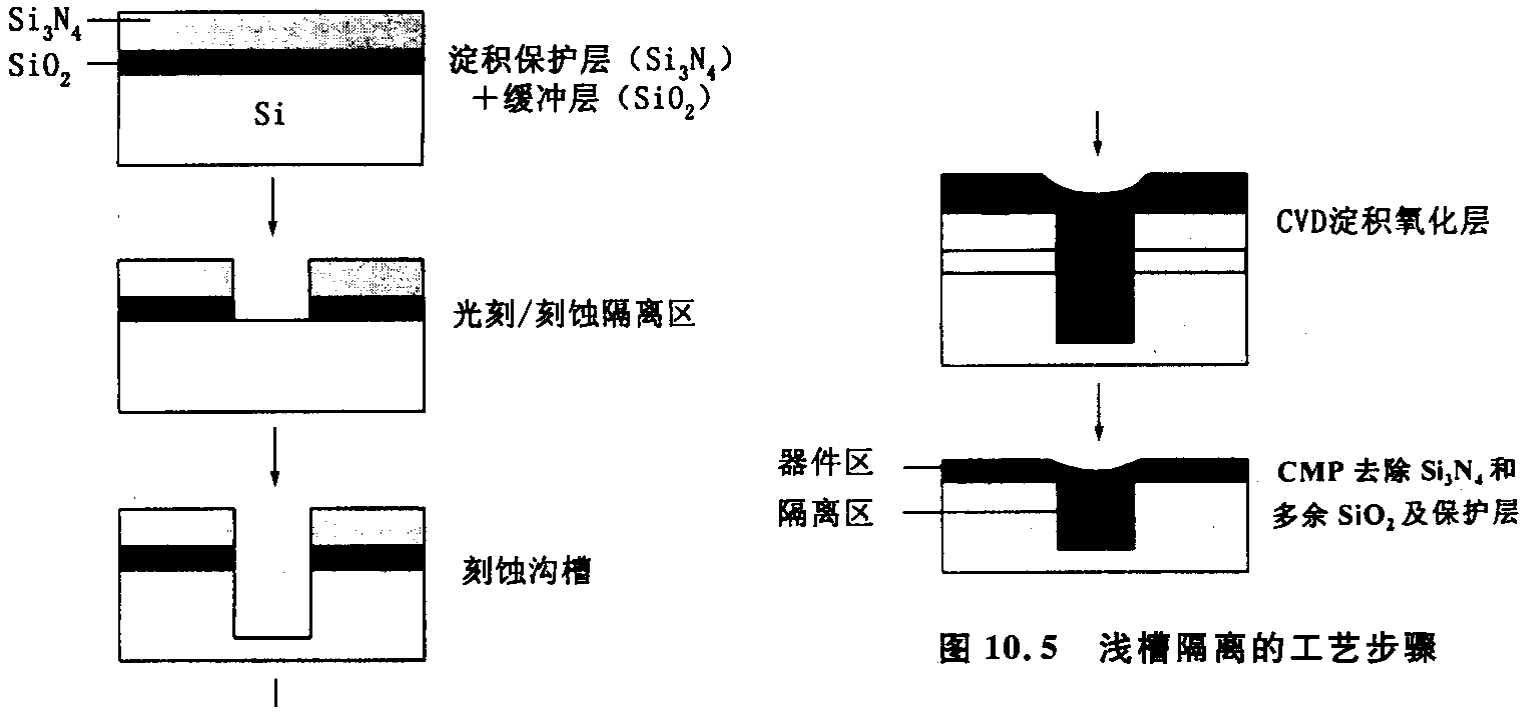


图 10.5　浅槽隔离的工艺步骤

通过氮化硅掩膜对硅基底进行图形化刻蚀，形成浅槽后填充氧化层以实现元件间的电学隔离，随后通过化学机械抛光(CMP)完成表面平坦化

Appendix: Threshold–Adjust Implant

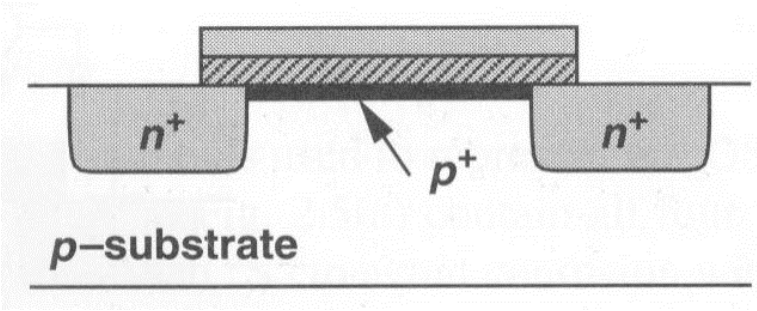


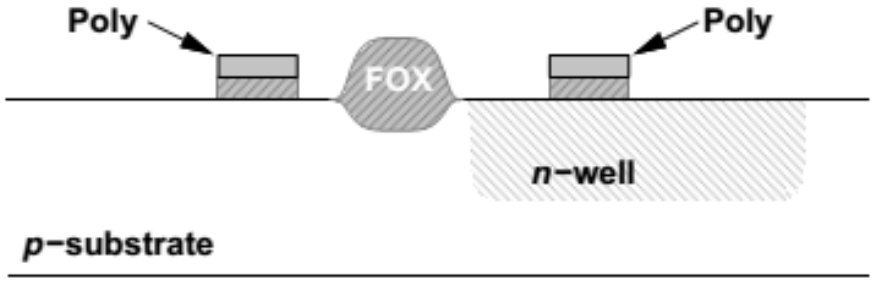
Figure 2.7 Implantation of p^+ dopants to alter the threshold.

$$V_{TH} = \Phi_{MS} + 2\Phi_F + \frac{Q_{dep}}{C_{ox}}$$

$$\Phi_F = (kT/q)\ln(N_{sub}/n_i)$$

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{\varepsilon_o \varepsilon_{sio_2}}{t_{ox}}$$

N-well Silicon-gate CMOS Fabrication (4)

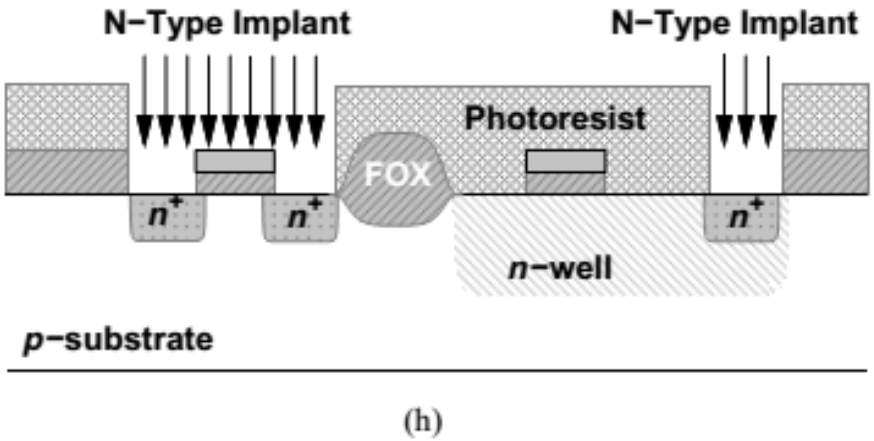


(g)

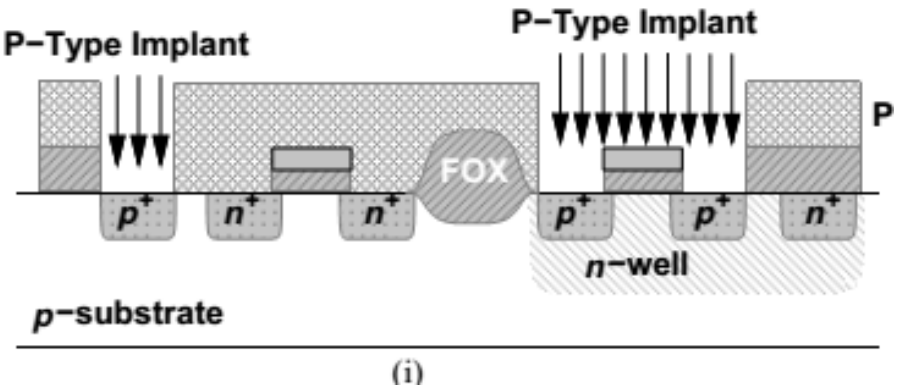
Poly mask

- With the gate oxide in place, the polysilicon layer is deposited
- Polysilicon is simply noncrystalline (amorphous) silicon
  -> conductor (additional implant)
- Self-aligned (自对准) Structure

N-well Silicon-gate CMOS Fabrication (5)



(h)



(i)

Source/drain mask + 2 lithography: source/drain junctions and the substrate and n-well ties are formed by ion implantation.

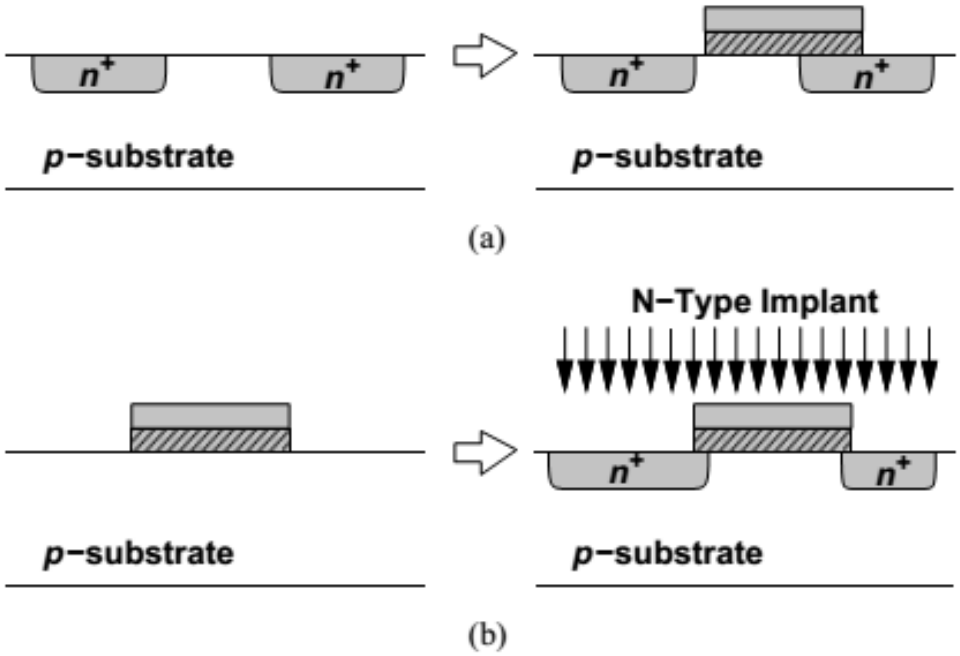
- negative photoresist
  expose the areas to implant n+ (the S/D junctions of NMOS and the n-well ties).
- positive photoresist
  expose the areas to implant p+ (the S/D junctions of PMOS and the substrate ties)

- Anneal and deposit a thick oxide (BPSG，硼磷硅玻璃)

These steps complete the fabrication of the basic transistors.

Front-End Processing (前道)

Appendix: Self-aligned (自对准) Structure



S/D junctions are formed after the gate oxide and polysilicon

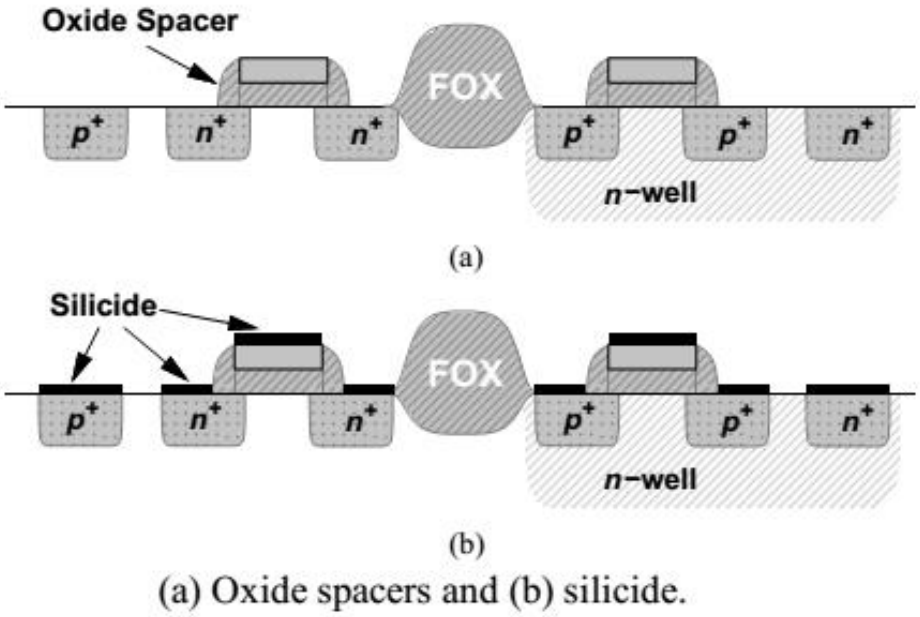
(a) Formation of n^+ regions before deposition of poly, (b) self-aligned structure.

Reason : Suppose, as Fig.(a), these junctions are created first. Then, the alignment of the gate poly mask with respect to the S/D areas becomes extremely critical.

N-well Silicon-gate CMOS Fabrication (6)

Back-End Processing（后道）: electrical connection on the chip through contacts and wires

Back-End Processing (1) -- Silicidation



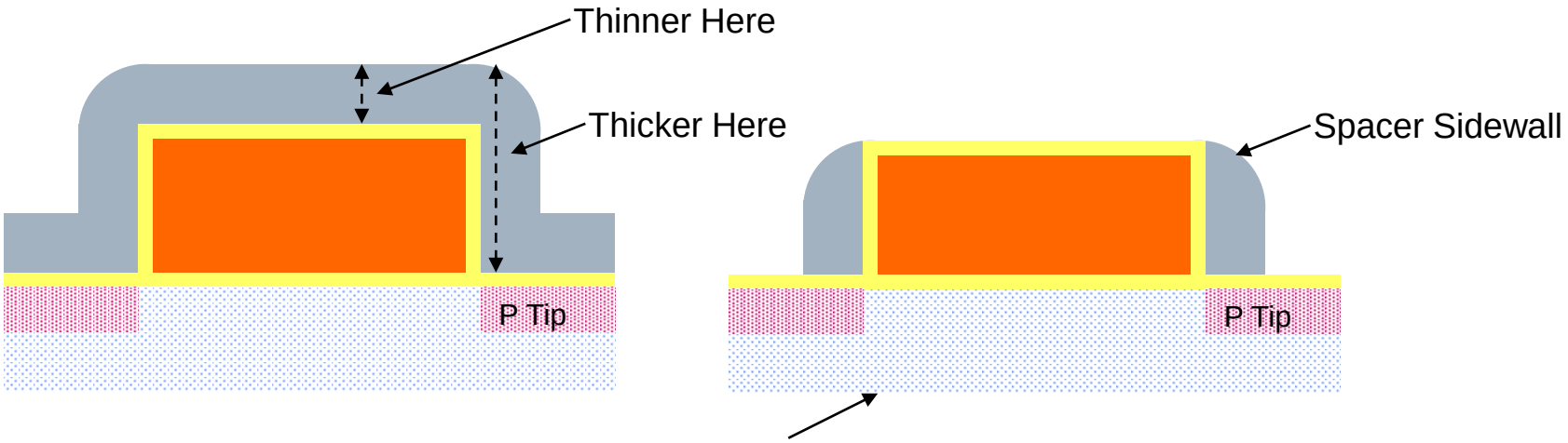
(a) Oxide spacers and (b) silicide.

-- **Oxide spacer**: at the edges of the poly gate such that the deposition of the silicide becomes a self-aligned process as well

-- $R_{\square}$ of poly and S/D：~10Ω/ □

Silicidation: reduce the resistance by covering the poly layer and active areas (S/D regions and sub and n-well ties) with a thin layer of a highly conductive material, e.g. $TiSi_2$、WSi_2、$TaSi_2$

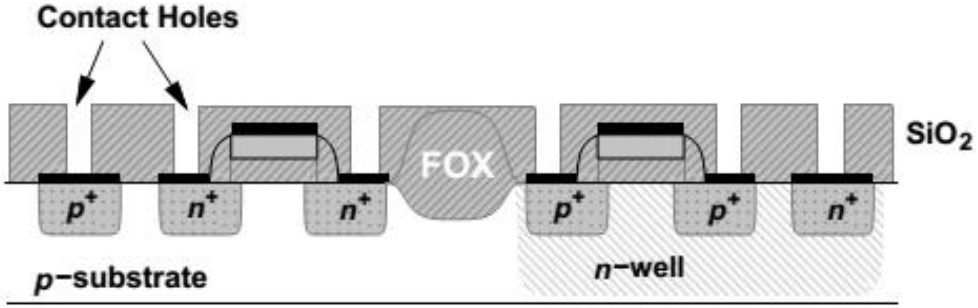
Appendix: Deposit Silicon Nitride Layer



Etch Nitride to Form Spacer Sidewalls: Using a carefully controlled etch, the thin nitride is removed from the horizontal surfaces, but the sidewalls remain. The sidewalls will precisely position the implants that form the transistor sources and drains.

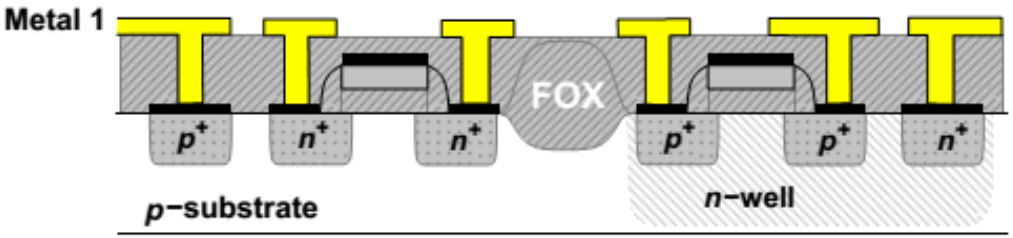
N-well Silicon-gate CMOS Fabrication (7)

Back-End Processing (2) – Contact and metal



Contact mask

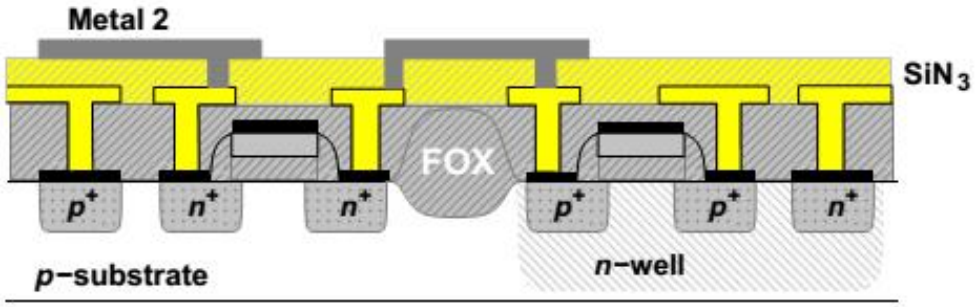
- cover the wafer with a relatively thick (0.3- to 0.5-μm) layer of oxide
- perform a lithography using the “contact mask”
- the contact holes are then created by plasma etching



Metal1 mask

- the first layer of metal interconnect (metal1) (Al or Cu) is deposited over the entire wafer
- a lithography using the “metal1 mask”
- the metal layer is selectively etched

N-well Silicon-gate CMOS Fabrication (8)



vias mask
metal2 mask

- the higher levels of interconnect are fabricated using the same procedure

- For each additional metal layer, two masks are required: one for the contact windows and another for the metal itself.

  Eg :A CMOS process having five layers of metal contains 10 masks for the back end.

- The contact windows between metal layers are sometimes called "**vias**" to distinguish them from the first level of contacts to active areas and polysilicon
- If a large area must be contacted, many small window (rather than a large window) are usually used.

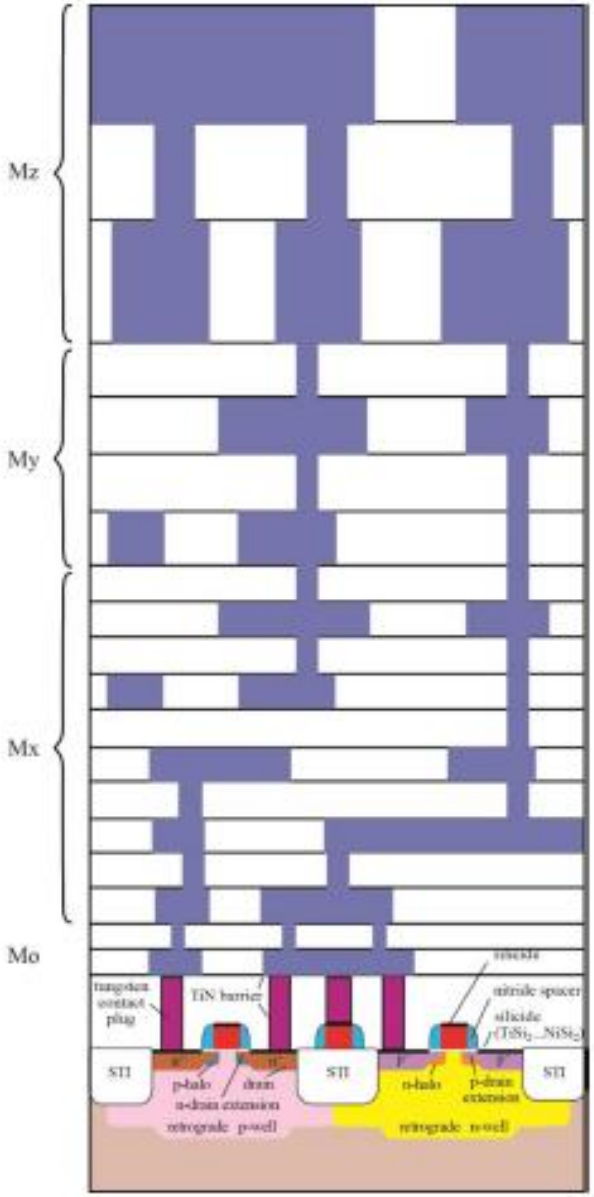
Appendix :Metal & Via

Metal

	METAL	Thick [µm]	W and S [µm]	R [Ω/mm]	C [fF/mm]
Mx	1	0.24	0.12	1k	54
My	2	0.31	0.14	500	35
My	3	0.31	0.14	500	28
My	4	0.31	0.14	500	24
My	5	0.31	0.14	500	22
Mz	6	0.85	0.42	52	22
Mz	7	0.85	0.42	52	22

VIA

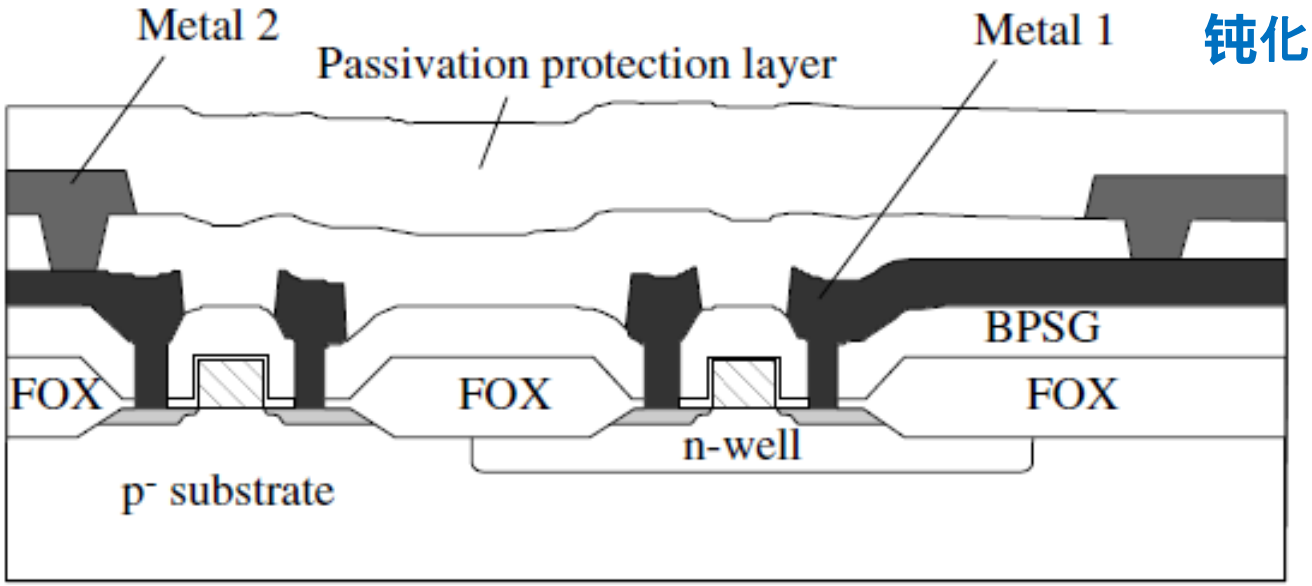
Contact type	between	Resistance (Ω)
CONTACT	POLY-METAL 1	18
VIA1-4	METALi-METALi+1	2.2
VIA5/6	METALj-METALj+1	0.7
VIA7	ALUCAP-METAL7	0.2



N-well Silicon-gate CMOS Fabrication (9)

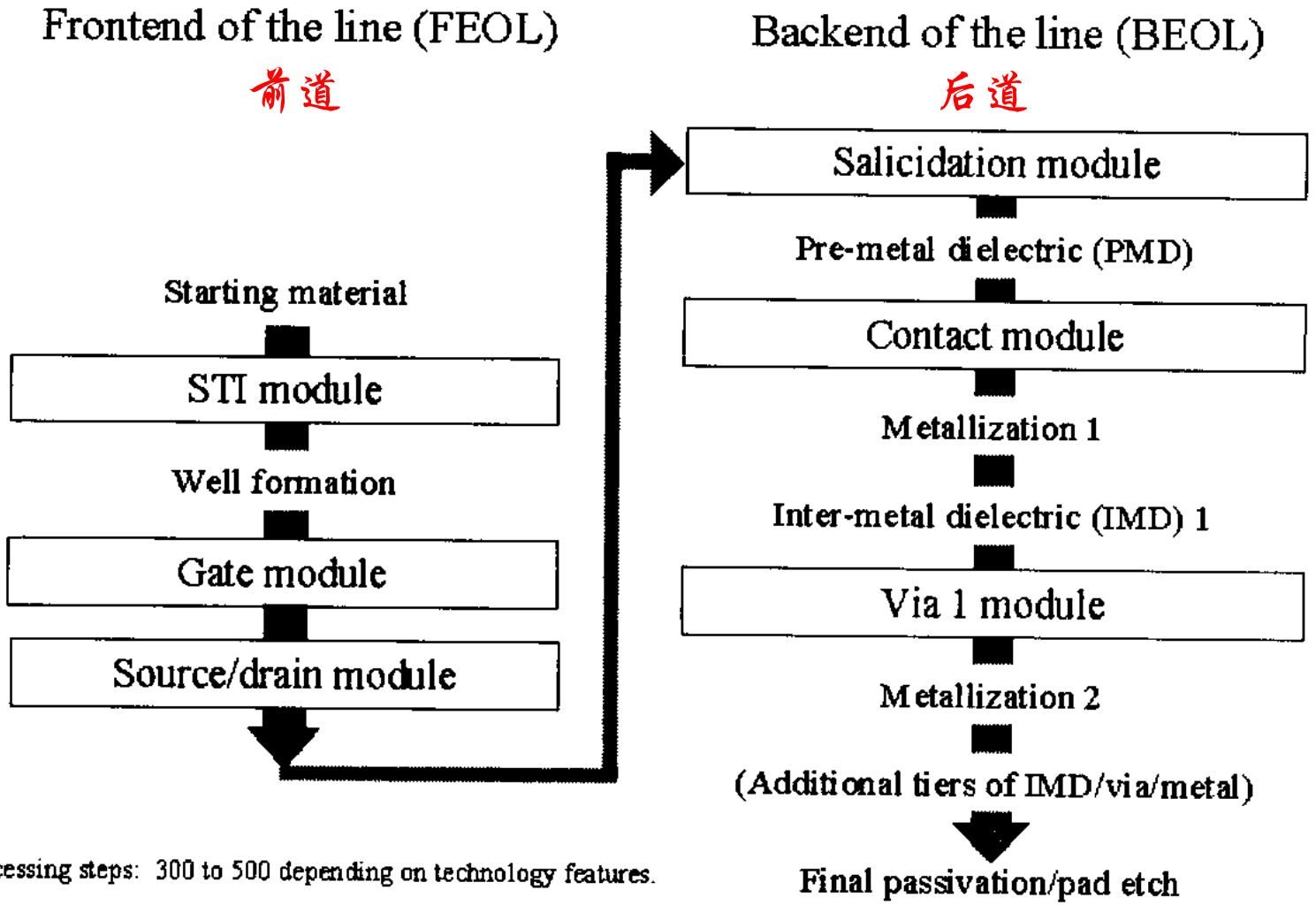
- Etch unwanted metal and deposit a **passivation layer** and open over **bonding pads**

Passivation mask
钝化层

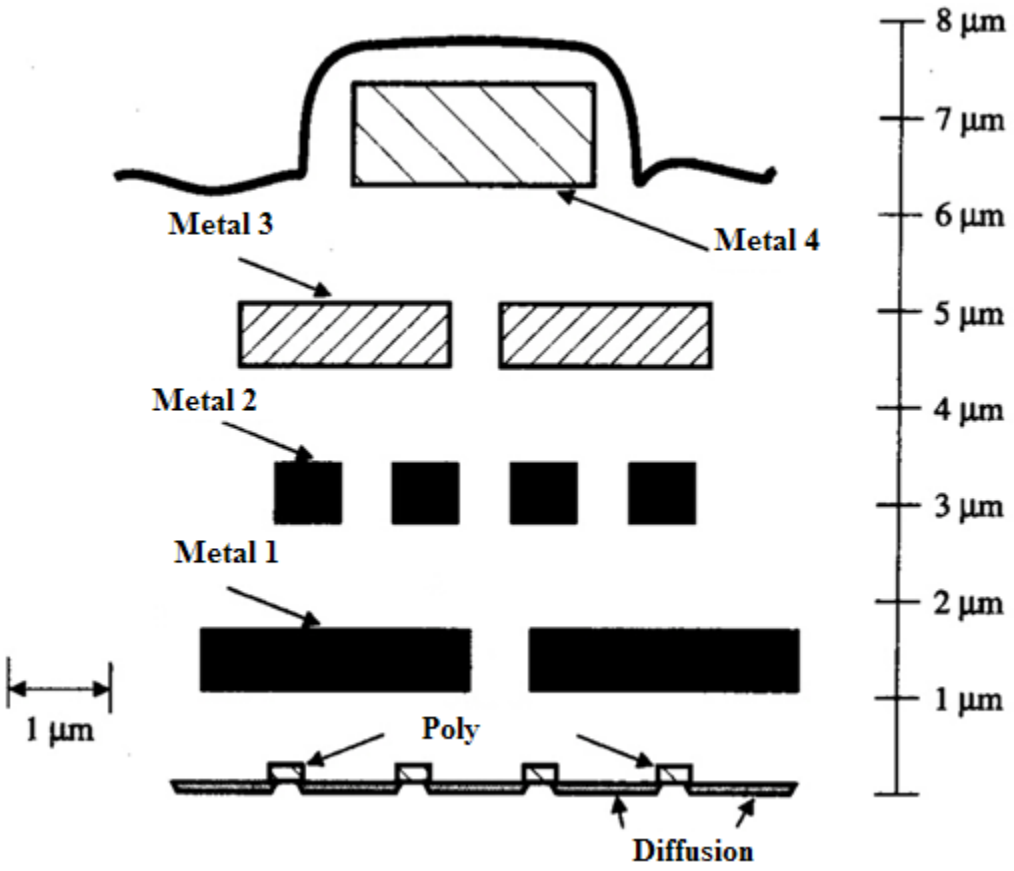


P-well process is similar but start with a p-well implant rather than an n-well implant.

FEOL vs. BEOL

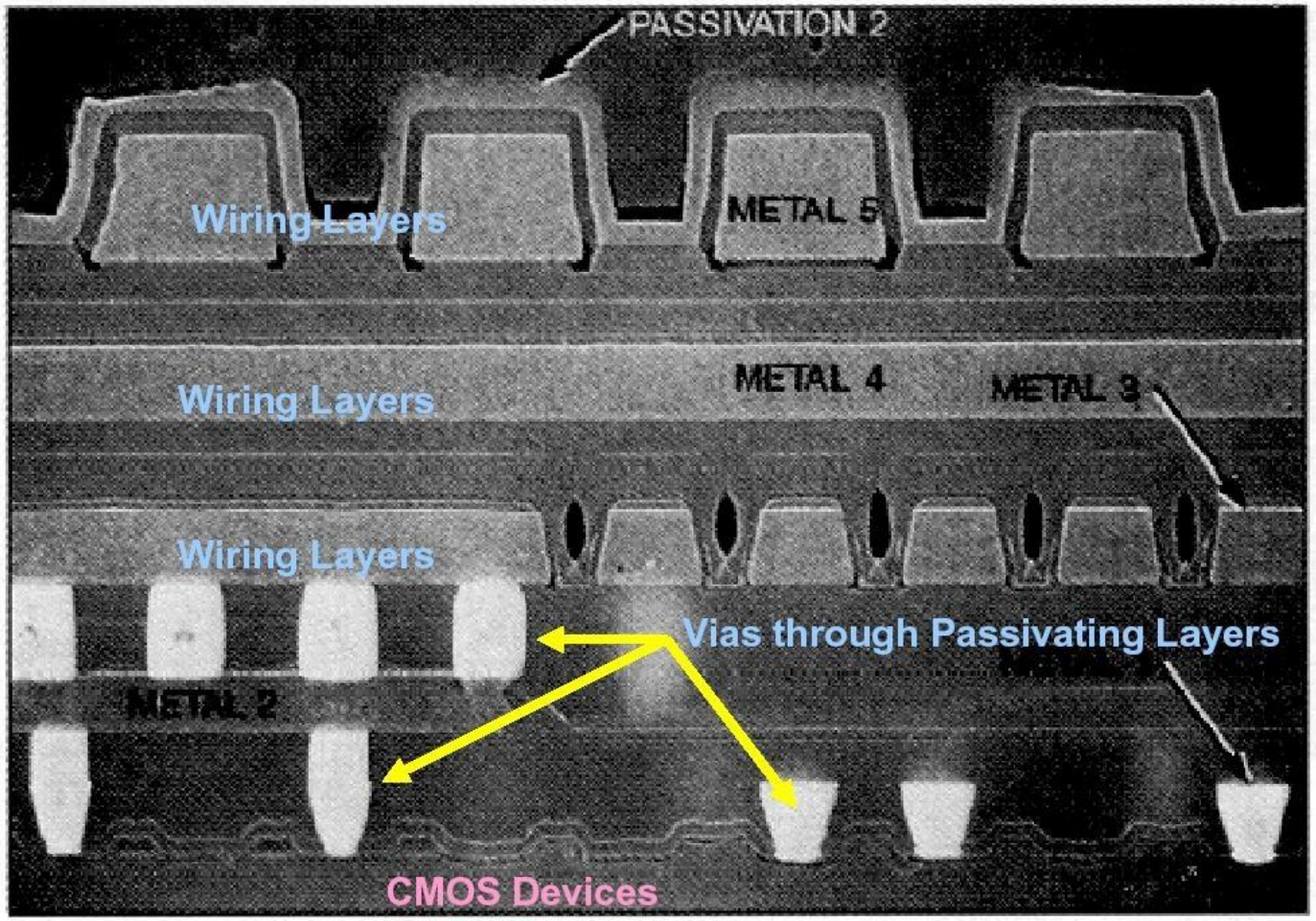

Frontend of the line (FEOL)
前道
Starting material
STI module
Well formation
Gate module
Source/drain module
Processing steps: 300 to 500 depending on technology features.
Backend of the line (BEOL)
后道
Salicidation module
Pre-metal dielectric (PMD)
Contact module
Metallization 1
Inter-metal dielectric (IMD) 1
Via 1 module
Metallization 2
(Additional tiers of IMD/via/metal)
Final passivation/pad etch

Cross-section of chip (1)

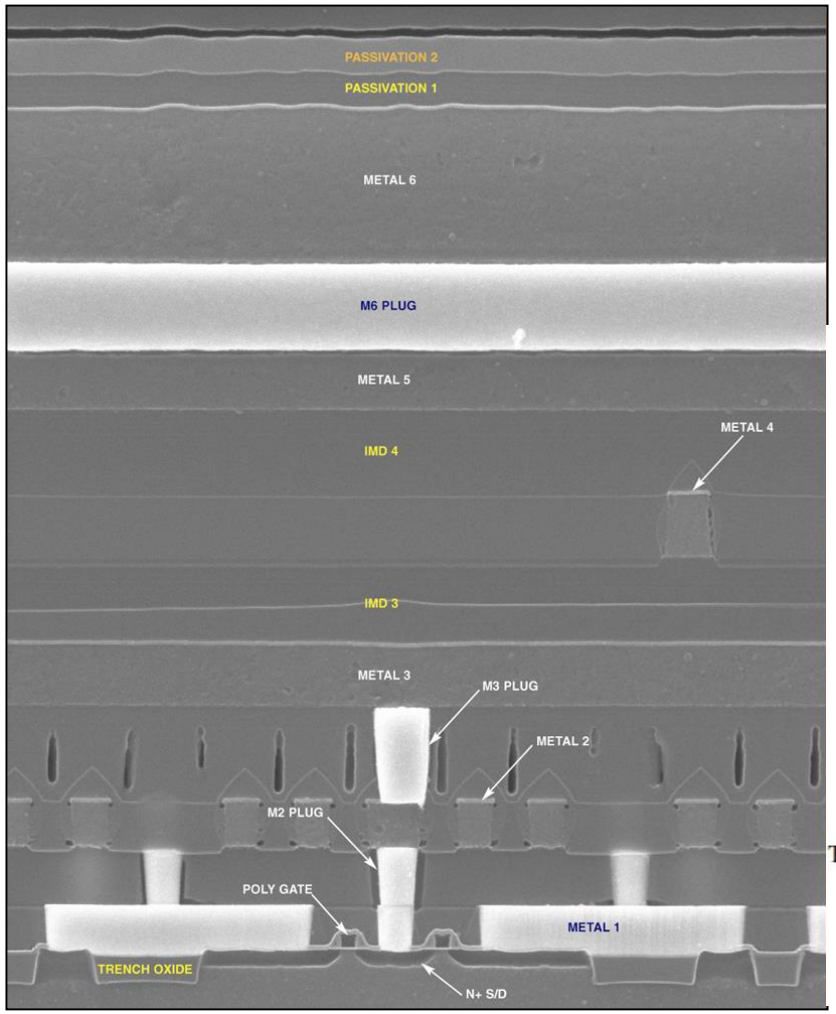


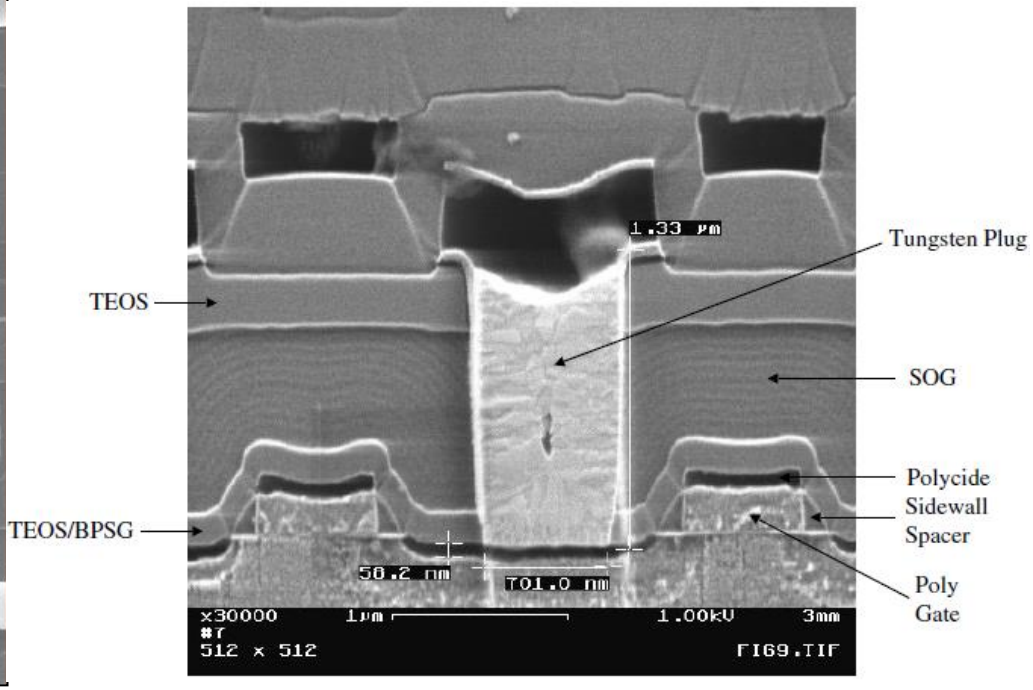
Cross-section of chip (2)

Scanning Electron Microscope

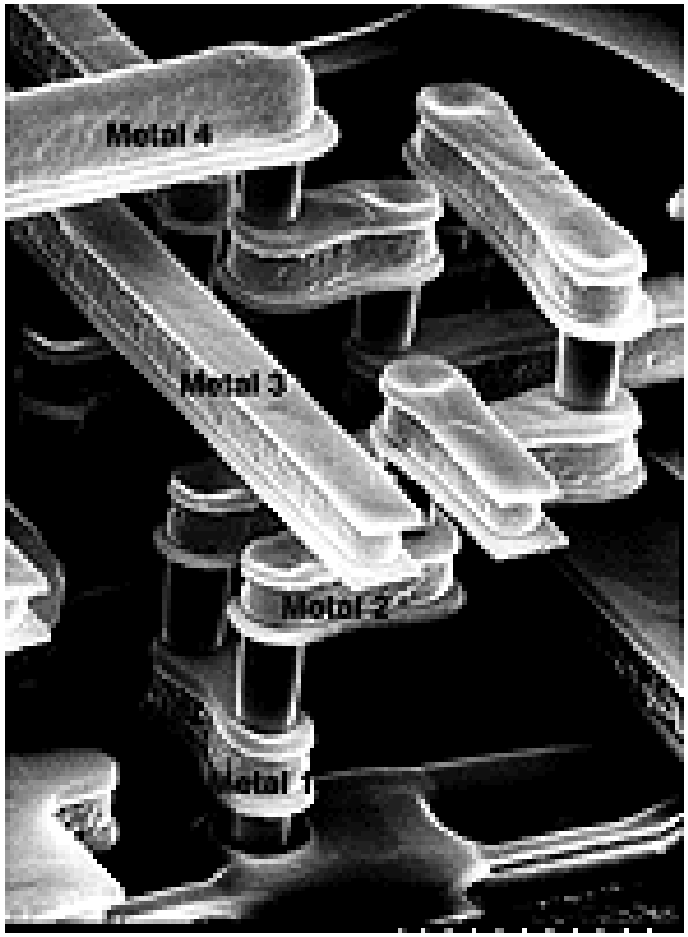


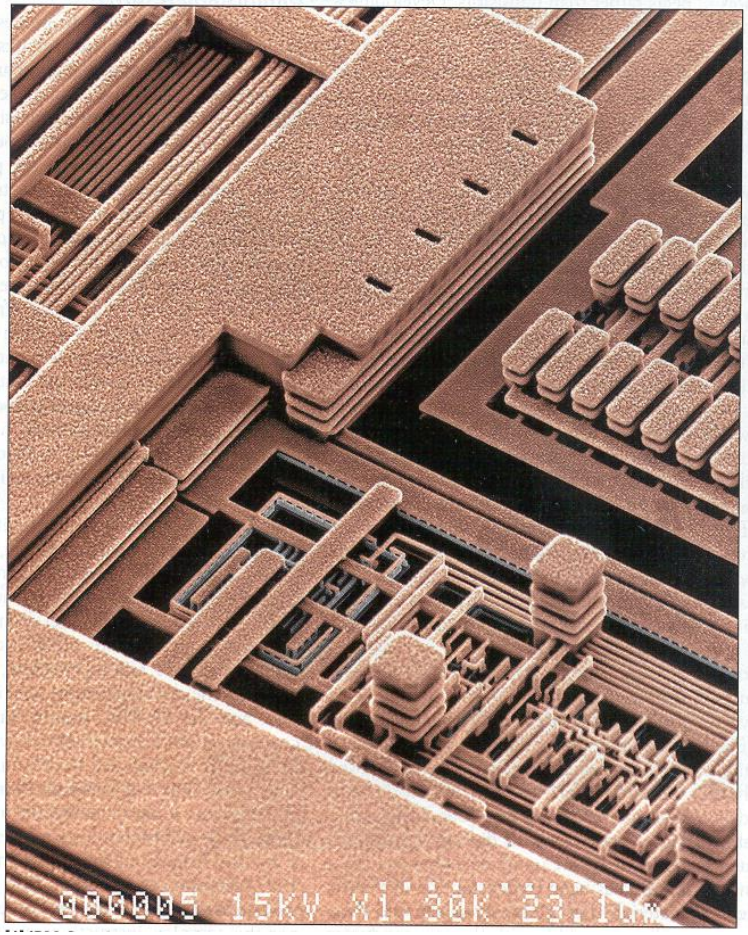
Cross-section of chip(3)





Advanced Metallization





Advanced CMOS process

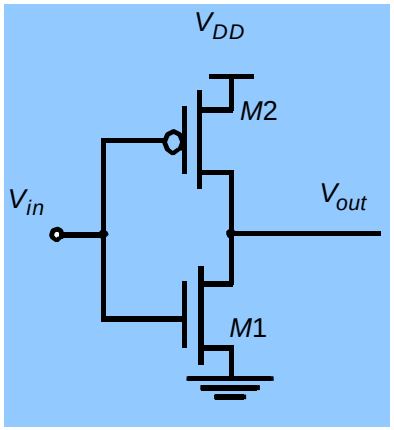
- 30 ~ 40 masks
- 6 ~12 metal interconnect layers
- Low-k intermetal dielectrics
  - ✓Less wire capacitance
  - ✓Faster and less power
- High-k gate dielectrics
  - ✓More transistor current
  - ✓Faster circuits
- Dual or triple threshold (V_{th}) process to support high-speed and low-power circuits
- ∘ ∘ ∘ ∘ ∘ ∘

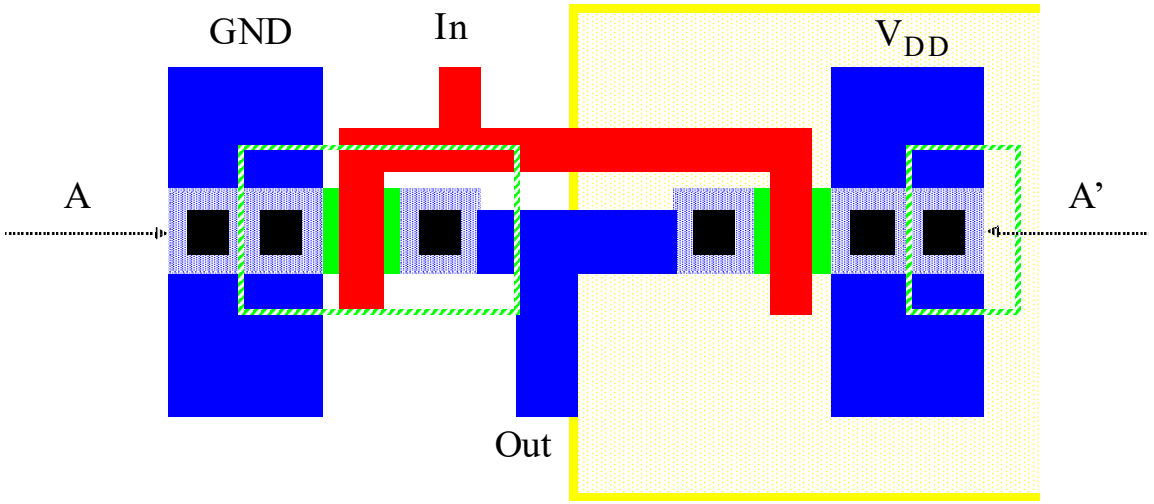
Outline

- Basic Semiconductor Fabrication Processes
- CMOS Fabrication Steps
- Layout

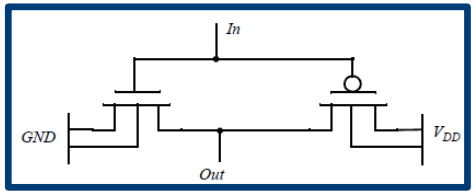
- Appendix: Design Rules

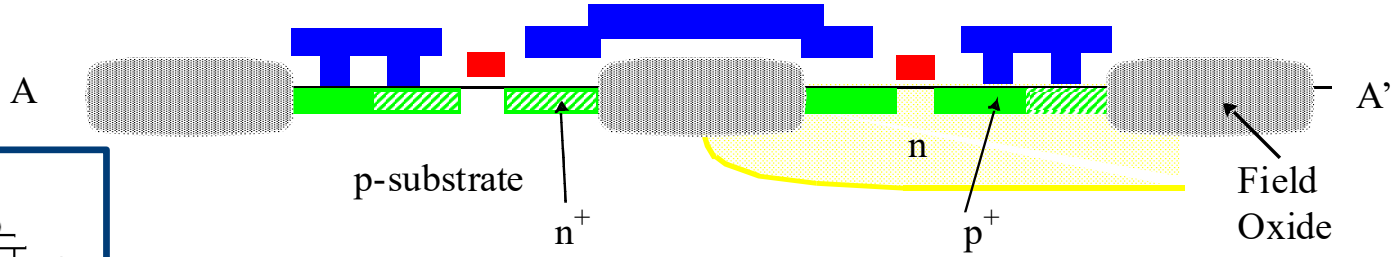
Layout Example: CMOS Inverter 1





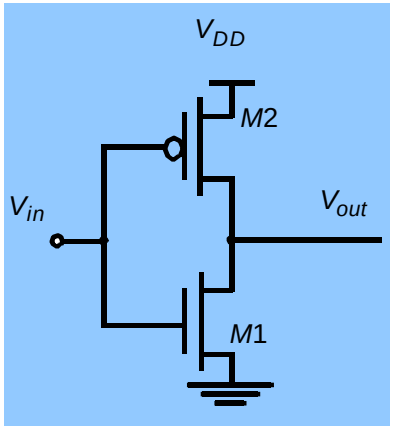
(a) Layout

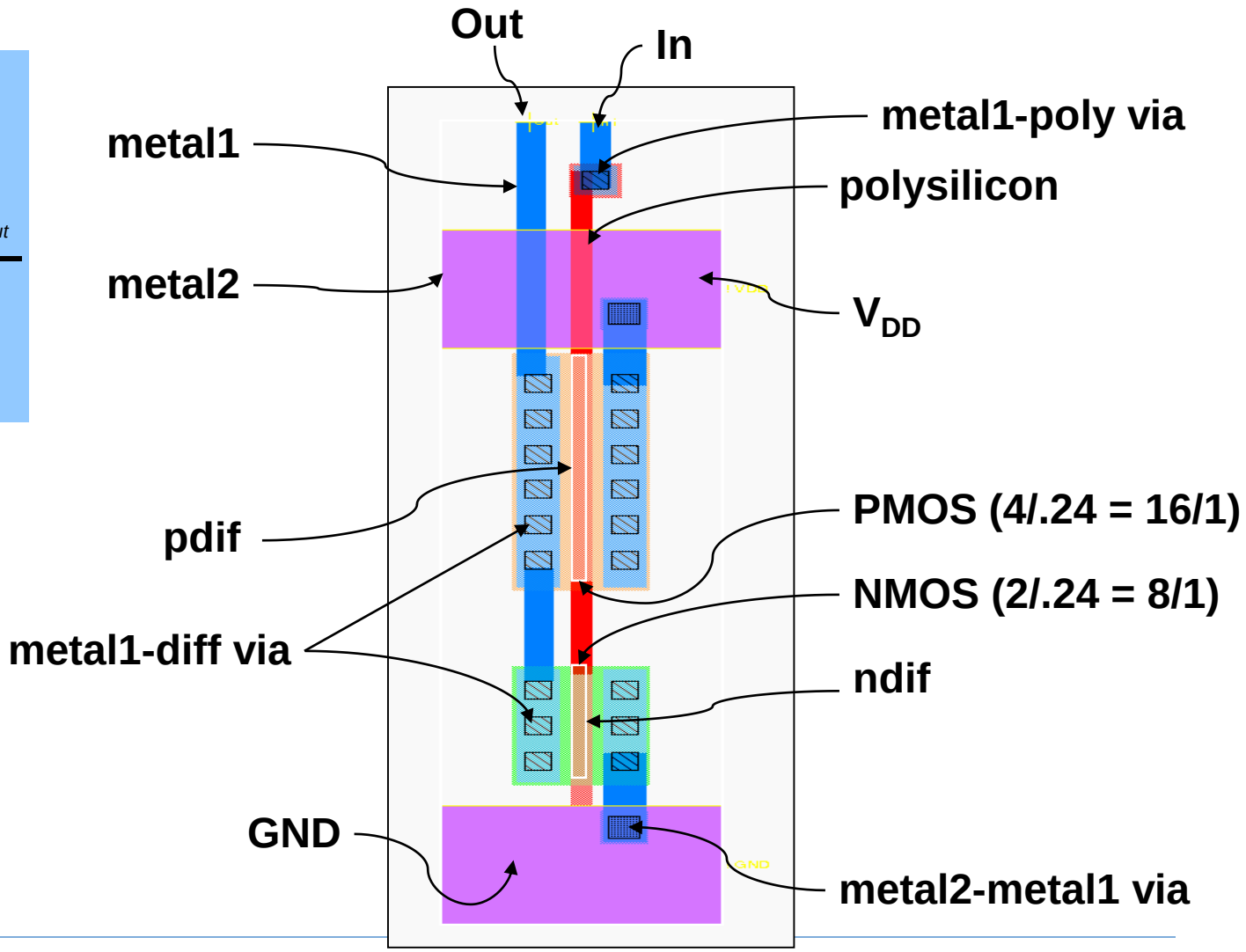




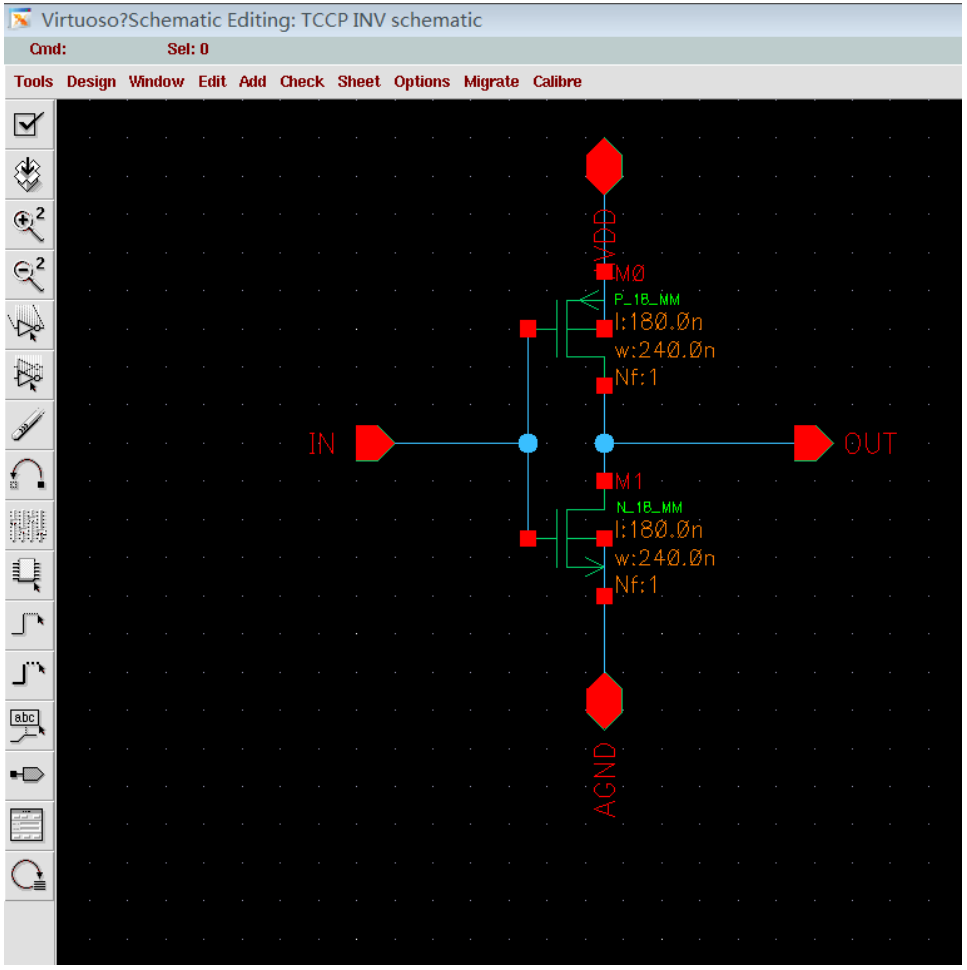
(b) Cross-Section along A-A'

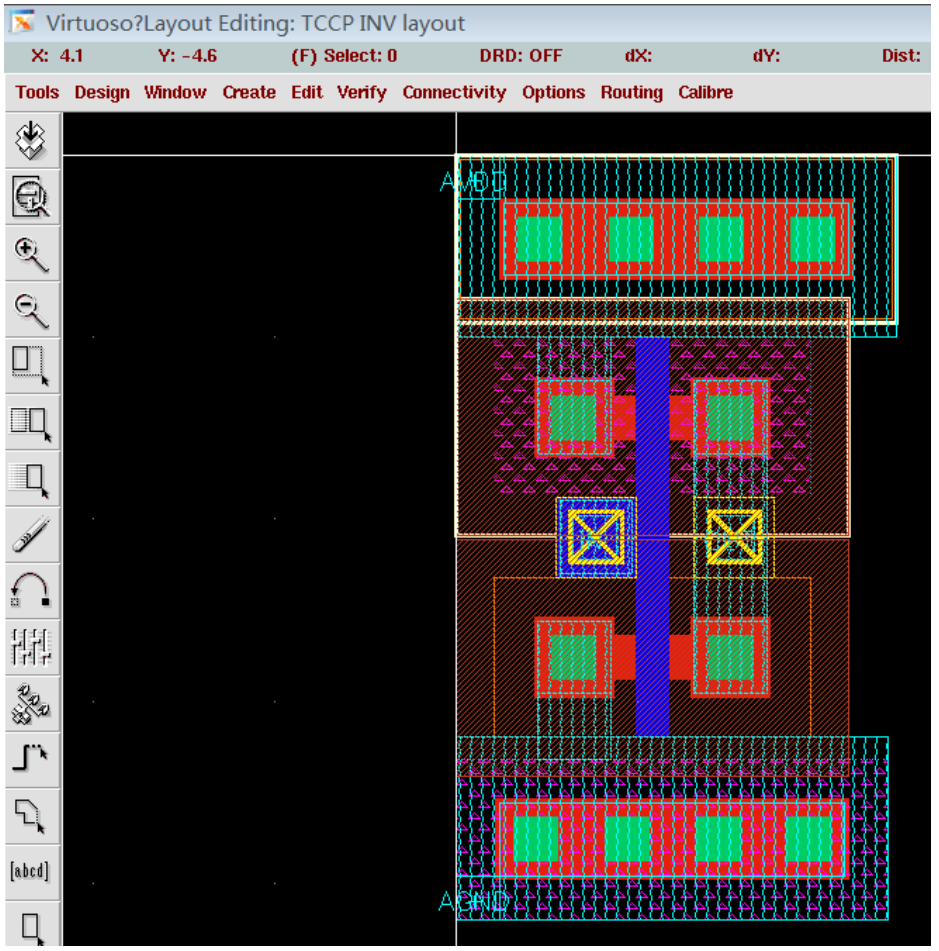
Layout Example: CMOS Inverter 2





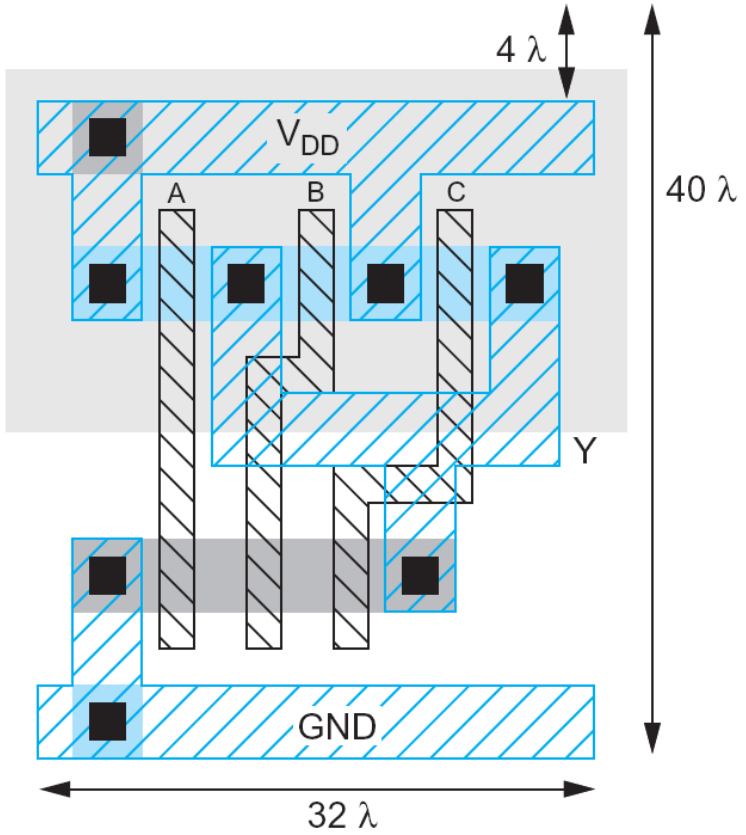
Layout Editor





Layout Example: CMOS NAND3

- Horizontal N-diffusion and p-diffusion strips
- Vertical polysilicon gates
- Metal1 V_{DD} rail at top
- Metal1 GND rail at bottom
- 32λ by 40λ



Appendix: Design Rules ***Page.61***

Summary

- **Manufacturing process of IC**
  - Require a large number of steps, each of which consists of a sequence of basic ***repetitively***
- **Optical masks**
  - The central interface between the intrinsic of manufacturing process and the design that the user wants to see transferred to the silicon fabric
- **Design rules set**
  - Define the constrains of minimum width and separation
  - Acts the contract between the circuit designer and the process engineer

集成电路原理与设计

2. 工艺流程

赵梦恋
zhaoml@zju.edu.cn

Appendix: Design Rules

- Layout: interface between designer and process engineer
- Rules constructed to ensure that design works even when small fab errors (within some tolerance) occur
  - Mask misalignment; Process parameters (e.g. lateral diffusion); Dust; Rough surface
- Guidelines for constructing process masks:
  - defines the geometries that appear on the masks used in fabrication
- A complete set includes
  - set of layers
  - intra-layer: relations between objects in the same layer
  - inter-layer: relations between objects on different layers

Design Rules(1)

- ***Minimum Width:*** the widths and lengths of geometries must exceed a minimum value imposed by lithography and processing capabilities



Fig. Excessive width variation in narrow poly line

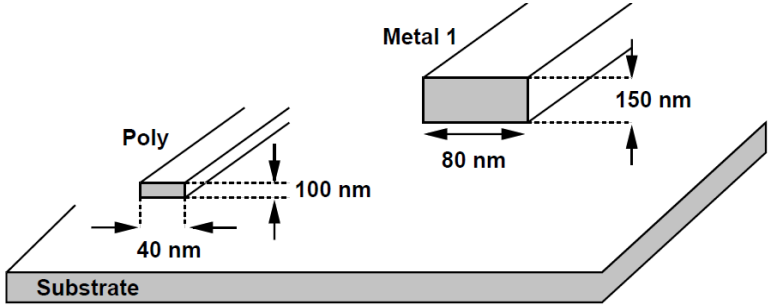
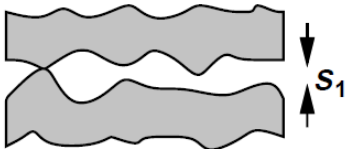
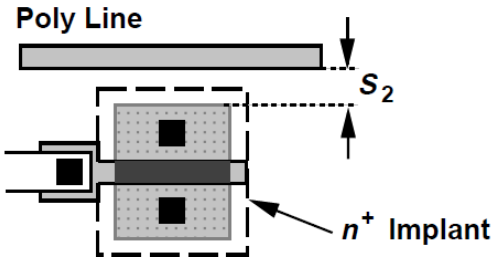


Fig. Widths and thicknesses of poly and metal lines

- ***Minimum Spacing :*** the geometries on the same mask or the different masks must be separated by a minimum spacing.





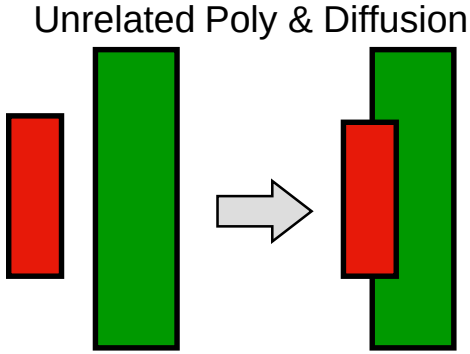


Fig. (a) Short between two excessively close poly lines, (b) minimum spacing between active and poly.

Design Rules(2)

- ***Minimum Enclosure:*** the sufficient margin of geometries surrounded by other masks
- ***Minimum Extension:*** some geometries mush extend beyond the edge of others by a minimum value

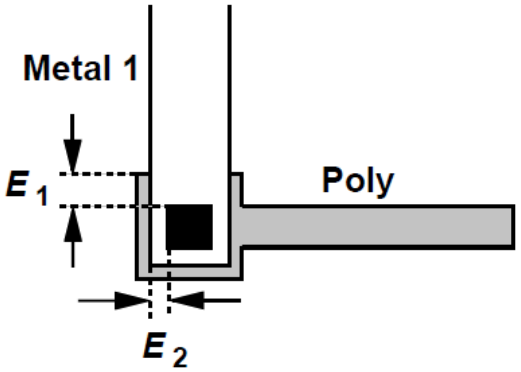


Fig. Enclosure rule for poly and metal surrounding a contact

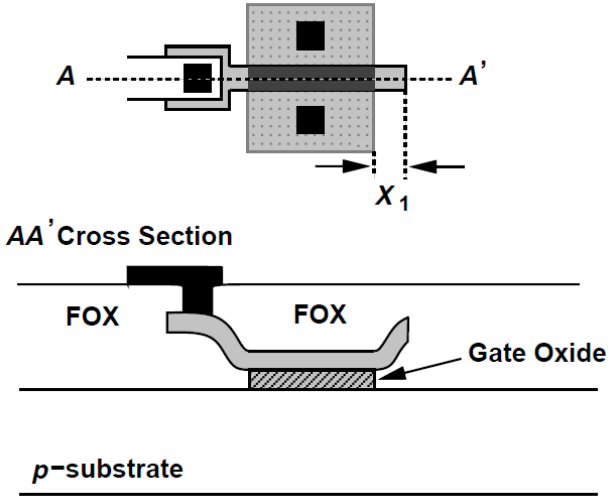
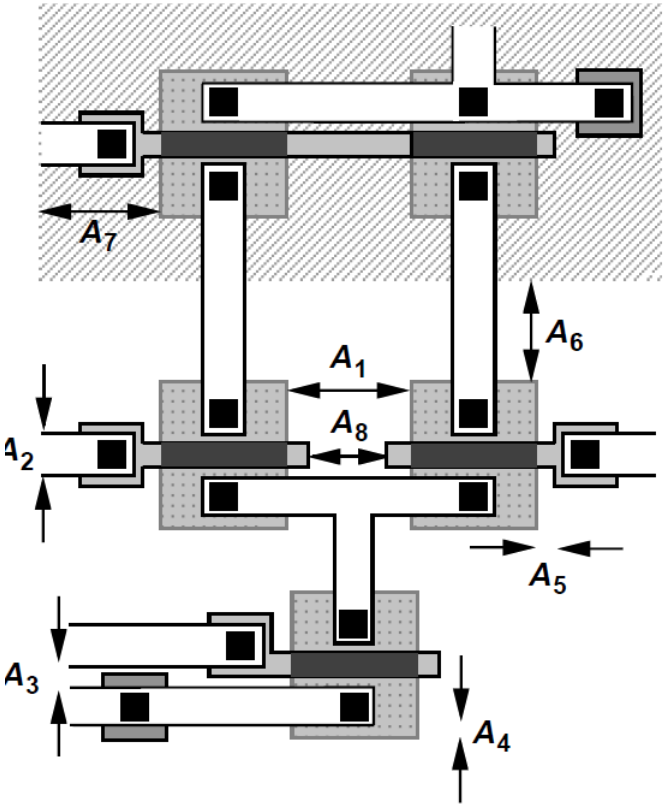


Fig. Extension of poly beyond the gate area

- ***Maximum Allowable Dimensions***

Design Rules(3)- Example

- ***Layout of a different pair with PMOS current source loads***



DRC: Design Rule Check
LVS: Layout vs. Schematics

A_1: Active-Active Spacing
A_2: Metal Width
A_3: Metal-Metal Spacing
A_4: Enclosure of Contact by Active
A_5: Poly-Active Spacing
A_6: Active-Well Spacing
A_7: Enclosure of Active by Well
A_8: Poly-Poly Spacing

Index of design Rule

* BEGIN *

0.18um Mixed-Mode/RFCMOS 1.8V/3.3V 1P6M Metal Metal Capacitor Process Topological Layout Rule

(Ver. 2.4_P1)

Single Poly 6 Metal (1P6M), Dual Voltage (1.8V & 3.3V)
P-Sub, TwiN_WELL/T_WELL CMOS Process
@ 1.8V/3.3Volt Operation

1. Contents

		Page
1.	Contents	1
2.	Revision History	3
3.	Mask Layer Definitions	9
4.	Layout Rules	
	4.1 DIFFUSION	10
	4.2 N_WELL	12
	4.3 T_WELL	13
	4.4 VTPL	14
	4.5 VTPHL	15
	4.6 P_WELL	16
	4.7 VTNL	16
	4.8 VTNI	17
	4.9 VTNHL	18
	4.10 VTN	19
	4.11 P-	19
	4.12 N-	19
	4.13 TG	20
	4.14 POLY1	21
	4.15 HR	23
	4.16 N+	25
	4.17 P+	26
	4.18 SAB and Non-Salicided resistors	27
	4.19 CONTACT	29
	4.20 METAL1	30
	4.21 MVIA1	31
	4.22 METAL2	32
	4.23 MVIA2	33
	4.24 METAL3	34
	4.25 MVIA3	35
	4.26 METAL4	36
	4.27 MVIA4	37
	4.28 METAL5	38
	4.29 METAL_CAP metal	39
	4.30 MVIA5	42

Design Rule of Metal 1

SPEC NO:G-03-MIXED_MODE/RFCMOS18-1.8V/3.3V-1P6M-MMC-TLR
Ver:2.4 Phase:1 Approved Date:06/22/2004

4.20 METAL1 Layer (No. P60) :

(unit : um)

20A.	Minimum METAL1 width	0.24
20B.	Minimum METAL1 to METAL1 spacing	
	a. METAL1 width < 10 um	0.24
	b. METAL1 width ≧ 10 um	0.28
20C.	Maximum METAL1 to METAL1 spacing	500
20D.	Minimum METAL1 overlap CONTACT	
	a. METAL1 line overlap CONTACT	0
	b. METAL1 line end overlap CONTACT	0.08
	c. METAL1 outer corner overlap CONTACT (for two adjacent sides, at least one side should be treated as metal line end)	0.08
	d. METAL1 inner corner overlap CONTACT	0
	e. METAL1 extension over CONTACT on opposite sides	0.08
20E.	Minimum METAL1 area	0.1764 um^2

20F. The METAL1 coverage must be larger than 30% of the entire chip area.

20G. Minimum METAL1 coverage for every millimeter square area is 25 %.

Note : It is strongly recommended that metal coverage should be evenly distributed in the whole chip; the more uniform, the better.

